SHARP RELAXATION, CUTOFF, AND GRAPH DIFFUSION FOR A LINEAR-POLYMER CHAIN

YITZCHAK SHMALO

Abstract. Consider the finite linear-polymer chain in which each pair of clusters merges at rate $2/n$ and each internal link breaks at rate β. We give a direct proof that its spectral gap equals the gap of the cluster-count chain, an equality announced by Granovsky and Erlihson, and prove the uniform estimate $\sqrt{\beta^2+4\beta}-5/n \le \gamma_M \le \sqrt{\beta^2+4\beta}+4/n$. Together with an all-parameter comparison and a critical scaling function at $\beta \asymp n^{-2}$, this determines the relaxation asymptotics throughout the parameter range. For $\beta = b/n$, the cluster-count chain has total-variation cutoff at $\sqrt{n}\log n/(8\sqrt{b})$ with a window of order at most $\sqrt{n}$. The same cutoff holds for the polymer chain started from one polymer or from all monomers. From arbitrary polymer configurations we prove a total-variation mixing bound of order $\sqrt{n}(\log n)^2$. Adding exclusion diffusion on any connected weighted graph gives the exact gap $\min\{\gamma_M, \eta_1 + D\lambda_G\}$, where λ_G is the one-particle graph gap and η_1 is a one-dimensional reaction eigenvalue for which we give uniform and sharp estimates. We also treat bounded inhomogeneous reactions, conservative nonreversible cut motion, and a stationary circulation that strictly accelerates the original polymer chain. The proofs use the stationary cycle representation, Boolean harmonic decomposition, positive eigenfunction comparisons, and hitting-time estimates.

1. Introduction

Consider a collection of linear polymers with total mass n. Each pair of polymers merges at a constant rate, and each internal link breaks independently. The resulting Markov chain has an explicit reversible law, but that law alone does not determine how fast the chain reaches equilibrium. Aldous [2, 3] asked for its relaxation scale when the fragmentation rate is comparable to $1/n$. We determine that scale with its leading constant, obtain a cutoff theorem, and study extensions with spatial motion and nonreversible rates.

The argument begins with the cycle representation in Aldous's notes. Its labelled cut process has more slow modes than the polymer chain. Rotation invariance removes the extra degree-one modes, and an explicit comparison puts every remaining harmonic sector above the first nonconstant count mode. This gives a direct proof of the equality between the polymer and count gaps announced by Granovsky and Erlihson [15]. A choice of reciprocal test function then gives a spectral estimate with an absolute error of order $1/n$.

1.1. The chain and the main theorem.

Fix $n \ge 2$ and $\beta > 0$. The state space is

$$\mathcal{S}_n = \left\{ m = (m_1, \dots, m_n) \in \mathbb{Z}_{\ge 0}^n : \sum_{j=1}^n j m_j = n \right\}.$$

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The manuscript, source code, and computational data are available at https://github.com/yspennstate/linear-polymer-relaxation.

Here m_j is the number of clusters containing j units of mass. Each cluster is viewed as a linear polymer with $j-1$ internal links. Every unordered pair of distinct cluster instances merges at rate $2/n$; the two instances may have the same size. Every internal link breaks at rate β.

Let e_j be the jth coordinate vector. The nonzero off-diagonal transition rates of the generator L_M are

$$q_M(m, m-e_j+e_i+e_{j-i}) = \begin{cases} 2\beta m_j, & 1 \le i < j/2, \\ \beta m_j, & i = j/2, \end{cases} \tag{1.1}$$

$$q_M(m, m+e_{i+j}-e_i-e_j) = \begin{cases} 2m_i m_j/n, & i<j, \quad i+j \le n, \\ m_i(m_i-1)/n, & i=j, \quad 2i \le n. \end{cases} \tag{1.2}$$

Rates with an inadmissible target are zero. The diagonal entries are the negatives of the row sums. The factors of two in (1.1) count the two distinct links giving the same unordered split when the two fragments have different sizes.

Write

$$K(m) = \sum_{j=1}^{n} m_j$$

for the number of clusters. The chain is irreducible and reversible with respect to

$$\pi_M(m) = \frac{1}{Z_M} \frac{(\beta n)^{K(m)}}{\prod_{j=1}^{n} m_j!}. \tag{1.3}$$

The normalizing constant Z_M is the sum of the displayed weights over $\mathcal{S}_n$. We rederive reversibility below, together with the comparison used in the proof.

For real functions on $\mathcal{S}_n$, let

$$\mathcal{E}_M(f,f) = -\langle f, L_M f\rangle_{\pi_M} = \frac{1}{2} \sum_{m,m' \in \mathcal{S}_n} \pi_M(m) q_M(m,m') \big(f(m') - f(m)\big)^2. \tag{1.4}$$

The spectral gap and relaxation time are

$$\gamma_M(n,\beta) = \inf_{\mathrm{Var}_{\pi_M} f > 0} \frac{\mathcal{E}_M(f,f)}{\mathrm{Var}_{\pi_M} f}, \qquad \tau_{\mathrm{rel}}(n,\beta) = \gamma_M(n,\beta)^{-1}. \tag{1.5}$$

We usually abbreviate $\gamma_M(n,\beta)$ to γ_M. All constants implicit in $\asymp$ below are independent of n and β.

Theorem 1.1. *For every $n \ge 2$ and $\beta > 0$, set*

$$s(n,\beta) = \beta + \sqrt{\beta} + \frac{1}{n}.$$

Then

$$\frac{s(n,\beta)}{6} \le \gamma_M(n,\beta) \le 4 s(n,\beta). \tag{1.6}$$

Consequently,

$$\frac{1}{4 s(n,\beta)} \le \tau_{\mathrm{rel}}(n,\beta) \le \frac{6}{s(n,\beta)}. \tag{1.7}$$

The proof also gives the estimates

$$\beta + \frac{2}{n} \le \gamma_M(n,\beta) \le \beta + 4 \max\left\{\sqrt{\beta}, \frac{1}{n}\right\}. \tag{1.8}$$

Theorem 1.1 remains useful at the transition $\beta \asymp n^{-2}$, where the additive error in the sharper estimate below is comparable to the gap. The one-state case $n = 1$ has no positive eigenvalue and is excluded throughout.

Corollary 1.2. *Let $\beta_n > 0$ and let $n \to \infty$.*

(1) *If $\beta_n \le Cn^{-2}$ for a fixed $C < \infty$, then $\tau_{\mathrm{rel}}(n, \beta_n) \asymp n$.*
(2) *If $n^{-2} \ll \beta_n \ll 1$, then $\tau_{\mathrm{rel}}(n, \beta_n) \asymp \beta_n^{-1/2}$.*
(3) *If $\beta_n \ge c > 0$, then $\tau_{\mathrm{rel}}(n, \beta_n) \asymp \beta_n^{-1}$.*

The comparison constants in the first and third assertions may depend on C and c, respectively. In particular, $\beta_n \asymp n^{-1}$ implies $\tau_{\mathrm{rel}}(n, \beta_n) \asymp \sqrt{n}$.

Proof. In each range, compare the three terms in $s(n, \beta_n)$ and apply (1.7). If $\beta_n \le Cn^{-2}$, the square-root term is at most $\sqrt{C}/n$. In the second range it dominates both other terms. If $\beta_n \ge c$, then $\sqrt{\beta_n} \le c^{-1/2}\beta_n$ and $n^{-1} \le (2c)^{-1}\beta_n$. □

For fixed β, Aldous's estimate is

$$(1 - o(1))(\beta^2 + 4\beta)^{-1/2} \le \tau_{\mathrm{rel}}(n, \beta) \le \beta^{-1} \qquad (n \to \infty). \tag{1.9}$$

His notes also discuss $\beta_n \sim b/n$, where the corresponding bounds have different powers of n [2]. The uniform theorem closes that order gap. The sharper results identify the constant in both regimes.

1.2. **Sharp conclusions and extensions.** Let γ_Q be the gap of the cluster-count chain. We prove

$$\gamma_M = \gamma_Q, \qquad \sqrt{\beta^2 + 4\beta} - \frac{5}{n} \le \gamma_M \le \sqrt{\beta^2 + 4\beta} + \frac{4}{n}$$

in Theorems 7.3 and 7.4. It follows that

$$\tau_{\mathrm{rel}}(n, b/n) \sim \frac{\sqrt{n}}{2\sqrt{b}}, \qquad \tau_{\mathrm{rel}}(n, \beta) \longrightarrow (\beta^2 + 4\beta)^{-1/2} \quad \text{for fixed } \beta > 0.$$

At the remaining transition scale we construct a positive, continuous function ρ such that

$$\left| n\gamma_M(n, b/n^2) - \rho(b) \right| \le b/n.$$

The function is the lowest eigenvalue of an explicit infinite birth–death operator. Finite tridiagonal matrices approximate it with a proved truncation error; see Section 8.

For $\beta = b/n$ with $b > 0$ fixed, the count process has worst-case total-variation cutoff at

$$t_n = \frac{\sqrt{n}\log n}{8\sqrt{b}},$$

with a window of order at most $\sqrt{n}$ (Theorem 9.1). The same statement holds for the polymer process from either extreme initial configuration: one polymer of size n, or n monomers. This transfer is exact at every time, because the density from either state remains a function of the cluster count. Separation cutoff from these states occurs at $2t_n$. For arbitrary initial partitions, Theorem 10.1 gives

$$t_{\mathrm{mix},n}^{\mathrm{polymer}}(\varepsilon) \le C_b\sqrt{n}\,[(\log n)^2 + \log(1/\varepsilon)].$$

Thus the worst-case polymer mixing time is determined within one logarithmic factor. This upper bound uses positivity of the actual polymer density and the degree-dependent sector energies.

The cycle representation also leads to a spatial process on cut sets. Add symmetric exclusion along a connected weighted graph G with diffusion coefficient D. Theorem 11.1 gives

$$\gamma_{\mathrm{sp}} = \min\{\gamma_Q, \eta_1 + D\lambda_G\},$$

where λ_G is the one-particle graph gap and η_1 is the lowest eigenvalue of the degree-one radial reaction operator. Writing $d = \sqrt{\beta^2 + 4\beta}$ and $c_1 = (\beta + d)/2$, we obtain

$$\min\{d, c_1 + D\lambda_G\} - \frac{5}{n} \le \gamma_{\mathrm{sp}} \le \min\{d, c_1 + D\lambda_G\} + \frac{4}{n}.$$

An additional critical scaling function describes η_1 when $\beta \asymp n^{-2}$. On a cycle the dynamics moves interval boundaries and merges or splits neighboring intervals. On a general graph it is an exclusion process with creation and deletion rates depending on the total occupancy.

Finally, bounded reversible changes of the reaction conductances give uniform estimates for inhomogeneous rates. Conservative cut motion generated by directed permutations has an exact formula for its asymptotic spectral rate and a separate formula for its L^2 contraction rate. A stationary current on four existing polymer configurations strictly increases the asymptotic spectral rate of the original partition-valued chain. These extensions are proved in Sections 11–13.

1.3. **Relation to earlier work.** Reversible clustering processes and their product-form equilibrium laws are classical; see Kelly [22, Chapter 8] and Whittle [27]. Durrett, Granovsky, and Gueron [13] characterize reversibility for a broad class of coagulation–fragmentation processes and study the resulting equilibrium distributions. Berestycki and Pitman [6] relate Gibbs laws on partitions to recursive fragmentation and its coalescent time reversal. The stationary law used here belongs to this established setting.

The solvable family of Granovsky and Erlihson [15] contains the present model. In their notation, set

$$N = n, \qquad a = 0, \qquad b = 2/n, \qquad \phi(1,1) = \beta.$$

Their Proposition 1 gives the autonomous count process, and their fragmentation formulas give the same unequal- and equal-fragment rates as (1.1). Their Section 2.3 announces equality of the full-process and count gaps and gives the bounds

$$\beta + 2/n \le \gamma_Q \le \beta + 2 - 2/n.$$

The transition formula displayed there concerns initial laws with Gibbs conditional distributions at fixed count. Our proof of the gap equality controls every orthogonal sector of every polymer observable, and the same comparison gives the uniform error estimate, the critical limit, and the graph extensions.

Hoze and Holcman [18] analyze the same constant-kernel model after a time change: take their fragmentation parameter to be $\beta n/2$ and multiply their generator by $2/n$. They obtain equilibrium cluster statistics and the mean times that two given particles spend together and apart. D'Orsogna, Lei, and Chou [12] study first assembly and equilibration in finite stochastic clustering systems, including restrictions on the largest allowed cluster. The all-observable spectral estimate and the mixing statements here concern the unrestricted partition-valued chain with the stated normalization.

Our one-dimensional arguments use positive-vector estimates for birth–death chains, developed in work including Zeifman [28] and Granovsky and Zeifman [16, 17]. The derivative identity is a finite unweighted instance of the intertwining relations of Chafaï and Joulin [10]. Filmus [14] gives the orthogonal harmonic structure on Boolean slices. We prove the finite identities used in the argument so that the cut-to-polymer comparison and its boundary conventions are explicit.

The cutoff step uses the birth–death hitting-time theorems of Chen and Saloff-Coste [11]. The calculation here supplies the endpoint means and variances for the moving equilibrium scale $K \asymp \sqrt{n}$. More general concentration and cutoff results for density-dependent chains appear in Barbour, Brightwell, and Luczak [4, 5]. A direct application of a fixed-density theorem would require checking both the n-dependent rescaled rates and the all-monomer starting state, whose rescaled density diverges. The passage-time proof handles these features directly.

Spatial coagulation–fragmentation has a separate literature. Bernardin and Toninelli [7] study a one-dimensional interval process with a dynamical phase transition and mixing estimates. Ioffe and Tóth [20] study split-and-merge limits for stationary random stirring on a lattice torus. Our graph model retains exchangeable reaction rates and adds exclusion motion. It is related to graph gap reductions for exclusion with reservoirs, including Salez [25, 26], and to the one- and two-particle identities of Kim and Sau [23]. The reservoir models have fixed sitewise creation and annihilation rates, whereas our deletion rate is $(|S|-1)/n$ and the reversible law is not a product law. Here the higher reaction sectors already dominate the count gap, leaving only the one-particle graph calculation in the exact formula. The proof therefore does not require the general interchange-process gap theorem of Caputo, Liggett, and Richthammer [9].

There are also established growing-kernel and nonreversible comparisons. Jiang [21] diagonalizes the Metropolis transposition chain with Ewens equilibrium. We record its spectral gap in the present time units as a known baseline for a different multiplicative coagulation rule. Antisymmetric perturbations that preserve equilibrium can accelerate convergence; Hwang, Hwang-Ma, and Sheu [19] give a spectral treatment for diffusions. Our four-state circulation verifies strict acceleration on the actual reaction graph of the linear-polymer chain.

Deterministic coagulation–fragmentation equations require different arguments. Aizenman and Bak [1] prove convergence through decay of free energy. Bhattacharya and Throm [8] prove convergence to equilibrium, with explicit rates, for continuous coagulation–fragmentation equations without detailed balance whose rate kernels are perturbations of constant kernels. The graph and kernel results in Sections 11–13 concern explicitly defined finite Markov generators and give no convergence theorem for general spatial coagulation–fragmentation equations.

1.4. **Structure of the proof.** There are two comparisons. First, the cluster count is a birth–death chain Q on $\{1,\dots,n\}$ with rates

$$\lambda_k = \beta(n-k), \qquad \mu_k = \frac{k(k-1)}{n}. \tag{1.10}$$

Every count observable is a polymer observable, so the variational principle gives $\gamma_M \le \gamma_Q$. Discrete differentiation turns the positive spectrum of $-Q$ into the spectrum of a killed birth–death operator. A positive function of the form $[(k+a)(k+a+1)]^{-1}$, with $a = \max\{1, n\sqrt{\beta}\}$, gives the upper bound in (1.8) directly.

For the lower bound, place n bonds in cyclic order and retain a nonempty set of cuts. Add each absent cut at rate β and remove each present cut at rate $(k-1)/n$ when there are k cuts. The unordered cyclic segment lengths define a polymer configuration. At stationarity this map preserves the variance and Dirichlet form of every lifted polymer observable. This is the precise comparison needed; the projected paths need not form a Markov chain from every deterministic cut configuration.

The cut-set generator commutes with all permutations of the bonds. Its harmonic decomposition reduces it to killed birth–death operators indexed by a degree ℓ. Degree zero is the count chain. Every polymer observable is invariant under cyclic rotation, which annihilates its degree-one component. The remaining sectors first give the uniform estimate and then, through a comparison with the positive count-derivative eigenfunction, the exact gap reduction. Reciprocal functions chosen by algebraic cancellation yield the sharp constants. Hitting-time means and variances give the cutoff theorem. For graph diffusion the harmonic sectors are tensor sums; all degrees above one already lie above the count gap before diffusion is added. A separate coupling proof of the square-root estimate is given in the appendix.

2. A stationary comparison with a labelled cut chain

Fix $n \ge 2$ and $\beta > 0$. Write L_M for the polymer generator in Theorem 1.1, and put $K(m) = \sum_j m_j$. We use a labelled auxiliary process, but only through its stationary Dirichlet form.

Arrange n labelled sites in cyclic order, with one bond from each site to the next, and let $[n]$ also index these bonds. For $n = 2$ these are two distinct bonds between the two sites. Thus the construction concerns a cyclic order and does not require a simple cycle graph. Let

$$\Omega_n = \{S \subseteq [n] : S \neq \varnothing\}.$$

The elements of S are the cut bonds. Define the generator

$$(L_B F)(S) = \beta \sum_{i \notin S} (F(S \cup \{i\}) - F(S)) + \frac{|S| - 1}{n} \sum_{i \in S} (F(S \setminus \{i\}) - F(S)). \tag{2.1}$$

When $|S| = 1$, the second sum is understood to be zero; the empty set is never a state. The process is irreducible: one can pass from S to T by adding bonds until reaching $S \cup T$ and then deleting the bonds outside T. Its reversible probability measure is

$$\pi_B(S) = \frac{1}{Z_B} \frac{(\beta n)^{|S|}}{(|S| - 1)!}. \tag{2.2}$$

Indeed, if $|S| = k$ and $i \notin S$, then $\pi_B(S \cup \{i\})/\pi_B(S) = \beta n/k$, whereas the reverse jump has rate k/n.

Cutting the cyclic order at the bonds in S produces $|S|$ segments. Let $p(S) = (m_j)_{j \geq 1}$ record their unordered lengths. In particular, $\sum_j j m_j = n$ and $K(p(S)) = |S|$.

Proposition 2.1. *The pushforward of π_B under p is*

$$\pi_M(m) = \frac{1}{Z_M} \frac{(\beta n)^{K(m)}}{\prod_j m_j!}. \tag{2.3}$$

It is reversible for L_M. For every function f on the polymer state space, the lift $F = f \circ p$ satisfies

$$\mathrm{Var}_{\pi_B}(F) = \mathrm{Var}_{\pi_M}(f), \qquad \mathcal{E}_B(F, F) = \mathcal{E}_M(f, f). \tag{2.4}$$

Consequently $\gamma_M \geq \gamma_B$. More generally, any Poincaré inequality for rotation-invariant functions on Ω_n holds, with the same constant, for polymer observables.

Proof. If m has k segments, the number of cut sets mapping to m is

$$N(m) = \frac{n(k-1)!}{\prod_j m_j!}. \tag{2.5}$$

To prove this formula, count cut sets together with a distinguished cut. On one hand there are $kN(m)$ such pairs. On the other hand the distinguished cut has n possible locations, and the ordered segment lengths have $k!/\prod_j m_j!$ possible arrangements. This also proves the formula when $k = 1$ or $k = n$. Multiplying (2.5) by the weight per cut set in (2.2) gives (2.3).

Conditional on $p(S) = m$, the cut set is uniform. Equivalently, after distinguishing a uniformly chosen cut, its segment lengths occur in uniform multiset order around the cycle. Adding an absent cut already has the required fragmentation rates in every configuration: in a segment of length j, the split into i and $j - i$ has two possible locations if $i < j/2$, and one if $i = j/2$. Thus its rate is $2\beta m_j$ or βm_j, respectively.

For mergers, suppose first that $k \geq 2$. At each of the k boundaries, the two neighbouring segment positions sample the multiset of lengths without replacement. For $i \neq j$, the expected number of boundaries between an i-segment and a j-segment is

$$k \left(\frac{m_i}{k} \frac{m_j}{k-1} + \frac{m_j}{k} \frac{m_i}{k-1} \right) = \frac{2 m_i m_j}{k-1}.$$

For equal sizes it is $m_i(m_i - 1)/(k - 1)$. A boundary is removed at rate $(k - 1)/n$, so the averaged merger rates are $2m_i m_j/n$ and $m_i(m_i - 1)/n$. These formulas include $k = 2$: the same two segments have two distinct boundaries, each with removal rate $1/n$. At $k = 1$ there is no merger.

These calculations prove the exact conditional identity

$$\mathbb{E}_{\pi_B}[L_B(f\circ p)(S)\mid p(S)=m]=(L_Mf)(m). \tag{2.6}$$

For two polymer functions f,g, reversibility of L_B and (2.6) imply

$$\langle f,L_Mg\rangle_{\pi_M}=\langle f\circ p,L_B(g\circ p)\rangle_{\pi_B}=\langle L_Mf,g\rangle_{\pi_M}.$$

Thus L_M is reversible under π_M. Taking $g=f$, and using $\mathcal{E}(f,f)=-\langle f,Lf\rangle$, proves the Dirichlet-form identity. The variance identity follows from the pushforward measure. Every polymer lift is invariant under a rotation of the labelled cyclic order, so the final assertions follow from the variational characterization of the spectral gap. □

Remark 2.2. Equation (2.6) is a stationary conditional average. The projection of the cut process need not be a Markov chain when started from an individual cut configuration. The proof uses only (2.4); no assertion of strong lumpability is needed.

3. Finite comparison and the cluster-count chain

We first record a positive-vector comparison for finite killed generators. Such comparisons are standard in the spectral analysis of birth–death processes; see, for example, [28]. We include the argument to specify exactly what is used.

Lemma 3.1. *Let I be a finite set and let*

$$(Ag)(x)=\sum_{y\neq x}r(x,y)(g(y)-g(x))-V(x)g(x),\qquad r(x,y)\geq 0,\quad V(x)\geq 0.$$

Suppose the off-diagonal graph is connected. Assume positive weights ρ satisfy $\rho(x)r(x,y)=\rho(y)r(y,x)$ for all x,y. If λ_* is the smallest eigenvalue of $-A$, then, for every strictly positive vector h,

$$\min_x\frac{-(Ah)(x)}{h(x)}\ \leq\ \lambda_*\ \leq\ \max_x\frac{-(Ah)(x)}{h(x)}. \tag{3.1}$$

The conclusion also holds on a one-point state space.

Proof. The operator is self-adjoint in the inner product weighted by ρ, and its quadratic form is

$$\langle g,-Ag\rangle_\rho=\frac12\sum_{x,y}\rho(x)r(x,y)(g(y)-g(x))^2+\sum_x\rho(x)V(x)g(x)^2.$$

A unit vector minimizing this form can be replaced by its absolute value without increasing the form. Hence a nonnegative minimizing eigenvector v exists. If $v(x)=0$, its eigenvalue equation implies $\sum_y r(x,y)v(y)=0$, so all its neighbours also have value zero. Connectedness shows that v is strictly positive.

At a point x where $v(x)/h(x)$ is minimal, the nonnegative off-diagonal entries give $(Av)(x)\geq(v(x)/h(x))(Ah)(x)$. Since $Av=-\lambda_*v$, this implies $\lambda_*\leq-(Ah)(x)/h(x)$. A point where v/h is maximal gives the reverse inequality. This proves (3.1). For one state the assertion is immediate. □

The number of clusters is itself a Markov coordinate of the polymer process. If it has value k, its total fragmentation and merger rates are, respectively, $\beta(n-k)$ and $k(k-1)/n$. Write Q for its generator on $\{1,\dots,n\}$:

$$(Qg)(k)=\beta(n-k)(g(k+1)-g(k))+\frac{k(k-1)}{n}(g(k-1)-g(k)). \tag{3.2}$$

The zero boundary coefficients suppress the nonexistent terms. The stationary measure is the pushforward of π_B under $S\mapsto|S|$. Testing the polymer variational principle with count observables gives

$$\gamma_M\leq\gamma_Q. \tag{3.3}$$

For $1 \le k \le n-1$, define the discrete derivative $(\partial g)(k) = g(k+1) - g(k)$. Direct expansion yields the intertwining

$$\partial Q = \mathcal{A}\partial, \tag{3.4}$$

where $\mathcal{A}$ is the killed birth–death generator

$$\begin{aligned}(\mathcal{A}h)(k) &= \beta(n-k-1)(h(k+1)-h(k)) + \frac{k(k-1)}{n}(h(k-1)-h(k)) \\ &\quad - \left(\beta + \frac{2k}{n}\right) h(k).\end{aligned} \tag{3.5}$$

This finite identity is an instance of the birth–death intertwinings studied in [10]. Here it can also be checked by comparing the three coefficients of $h(k-1), h(k), h(k+1)$. The derivative is an isomorphism from functions modulo constants to $\mathbb{R}^{n-1}$. Therefore the eigenvalues of $-\mathcal{A}$ are exactly the nonzero eigenvalues of $-Q$, with multiplicity. In particular its smallest eigenvalue is γ_Q.

Every finite birth–death matrix with positive interior rates is reversible under positive weights, obtained recursively from detailed balance. Lemma 3.1 thus applies to $\mathcal{A}$; for $n = 2$ it is a one-point matrix. The choice $h = 1$ gives

$$\gamma_Q \ge \beta + \frac{2}{n}. \tag{3.6}$$

Proposition 3.2. *For every $n \ge 2$ and $\beta > 0$,*

$$\gamma_M \le \gamma_Q \le \beta + 4\max\{\sqrt{\beta}, 1/n\} \le 4\left(\beta + \sqrt{\beta} + \frac{1}{n}\right). \tag{3.7}$$

Proof. Put $a = \max\{1, n\sqrt{\beta}\}$ and choose $h(k) = ((k+a)(k+a+1))^{-1}$. The adjacent ratios are

$$\frac{h(k+1)}{h(k)} = \frac{k+a}{k+a+2}, \qquad \frac{h(k-1)}{h(k)} = \frac{k+a+1}{k+a-1}.$$

Substitution into (3.5) gives the exact identity

$$\frac{-(\mathcal{A}h)(k)}{h(k)} = \frac{2\beta(n-k-1)}{k+a+2} + \beta + \frac{2ka}{n(k+a-1)} \le \frac{2\beta n}{a} + \beta + \frac{2a}{n}. \tag{3.8}$$

The last inequality uses $a \ge 1$. If $a = n\sqrt{\beta}$, the last expression is $\beta + 4\sqrt{\beta}$. If $a = 1$, then $\beta n \le 1/n$ and it is at most $\beta + 4/n$. Apply the upper half of Lemma 3.1, followed by (3.3). □

4. Harmonic decomposition and rotation-invariant observables

The cut chain has the full permutation symmetry of $[n]$. Its degree-one modes can be slow when β is very small, but those modes do not occur in polymer observables. We give the relevant Boolean-lattice decomposition explicitly. Related harmonic descriptions of functions on slices of the Boolean cube appear in [14].

For $0 \le j \le n$, let V_j be the space of real functions on the j-subsets of $[n]$, with counting inner product. Set $V_{-1} = V_{n+1} = \{0\}$. Define the upward and downward maps by

$$(Ua)(T) = \sum_{i \in T} a(T \setminus \{i\}), \qquad (Da)(R) = \sum_{i \notin R} a(R \cup \{i\}).$$

Thus $U : V_j \to V_{j+1}$ and $D : V_j \to V_{j-1}$ are adjoints on adjacent layers. On V_j, direct double counting gives

$$DU - UD = (n-2j)I. \tag{4.1}$$

Indeed, the diagonal term of DU counts $n-j$ extensions and that of UD counts j deletions. Every other term exchanges one present and one absent label and occurs once on each side.

For $0 \le \ell \le \lfloor n/2 \rfloor$, set $H_\ell = \ker(D : V_\ell \to V_{\ell-1})$. If $a \in H_\ell$, define its lift on a layer of k-subsets by

$$F_a(S) = \sum_{\substack{T \subseteq S \\ |T|=\ell}} a(T) = \frac{U^{k-\ell} a(S)}{(k-\ell)!} \quad (k \ge \ell), \tag{4.2}$$

and put $F_a = 0$ when $k < \ell$.

Lemma 4.1. *For every $0 \le k \le n$, the layer V_k is the orthogonal direct sum of the lifted spaces H_ℓ, with $0 \le \ell \le \min\{k, n-k\}$. Their dimensions and lift norms are*

$$\dim H_\ell = \binom{n}{\ell} - \binom{n}{\ell-1}, \qquad \|F_a\|_{V_k}^2 = \binom{n-2\ell}{k-\ell} \|a\|_{V_\ell}^2, \tag{4.3}$$

where $\binom{n}{-1} = 0$. The lift is injective for $\ell \le k \le n-\ell$ and vanishes for $k > n-\ell$.

Proof. Taking an inner product in (4.1) gives

$$\|Ua\|^2 = \|Da\|^2 + (n-2j)\|a\|^2 \qquad (a \in V_j).$$

Hence U is injective for $j < n/2$, and its adjoint on the next layer is surjective. This proves the dimension formula for H_ℓ.

For $a \in H_\ell$, induction using the commutator gives

$$DU^j a = j(n-2\ell-j+1)U^{j-1}a. \tag{4.4}$$

For the induction step, commute D past the last U and use

$$(j-1)(n-2\ell-j+2) + n - 2(\ell+j-1) = j(n-2\ell-j+1).$$

Consequently

$$\|U^j a\|^2 = j(n-2\ell-j+1)\|U^{j-1}a\|^2 = \frac{j!(n-2\ell)!}{(n-2\ell-j)!}\|a\|^2$$

for $0 \le j \le n-2\ell$. Dividing by $(j!)^2$ proves the norm formula and injectivity. At $j = n-2\ell+1$ the preceding norm recursion gives zero, so all subsequent lifts vanish.

For orthogonality, let $a \in H_\ell$, $b \in H_h$, and $\ell < h$, with both lifts supported on layer k. Move the $k-h$ upward maps in the lift of b across the inner product, and apply (4.4). The resulting inner product is a scalar multiple of $\langle U^{h-\ell}a, b\rangle$, which is zero because $Db = 0$. Finally the dimensions telescope:

$$\sum_{\ell=0}^{\min(k,n-k)} \left(\binom{n}{\ell} - \binom{n}{\ell-1} \right) = \binom{n}{\min(k, n-k)} = \binom{n}{k}.$$

Thus the orthogonal subspaces exhaust V_k. □

On the nonempty Boolean lattice, the degree-ℓ sector consists of all functions obtained by multiplying the lifts of a basis of H_ℓ by arbitrary functions of the layer k. For degree zero the layers are $1, \dots, n$; for degree $\ell \ge 1$ they are $\ell, \dots, n-\ell$. The layerwise decomposition in Lemma 4.1 is orthogonal also in $L^2(\pi_B)$, because π_B is constant on each layer.

Lemma 4.2. *Each degree sector is preserved by L_B. Degree zero is the count generator Q. For $\ell \ge 1$, each harmonic direction has radial generator*

$$\begin{aligned}(\mathcal{L}_\ell g)(k) &= \beta(n-k-\ell)(g(k+1) - g(k)) \\ &\quad + \frac{(k-1)(k-\ell)}{n}(g(k-1) - g(k)) - \left(\ell\beta + \frac{\ell(k-1)}{n}\right) g(k),\end{aligned} \tag{4.5}$$

on $\ell \le k \le n-\ell$. It is a reversible killed birth–death generator. The spectrum of $-L_B$ is the union of the spectrum of $-Q$ and the spectra of $-\mathcal{L}_\ell$, the latter repeated $\dim H_\ell$ times.

Proof. The lift satisfies

$$\sum_{i\in S} F_a(S\setminus\{i\}) = (k-\ell)F_a(S), \qquad \sum_{i\notin S} F_a(S\cup\{i\}) = (n-k-\ell)F_a(S). \tag{4.6}$$

The first identity counts each ℓ-subset of S exactly $k-\ell$ times. For the second, the terms already contained in S contribute $(n-k)F_a(S)$. The remaining terms can be grouped as

$$\sum_{\substack{R\subseteq S\\ |R|=\ell-1}} \sum_{i\notin S} a(R\cup\{i\}).$$

By $Da=0$, the inner sum is the negative of the corresponding sum over $i\in S\setminus R$. Summing over R counts each ℓ-subset ℓ times, giving $-\ell F_a(S)$. For $\ell=0$ the identities hold directly.

Substitute (4.6) in (2.1) to obtain

$$L_B(F_a g)(S) = F_a(S)(\mathcal{L}_\ell g)(|S|).$$

The same identities, with the zero lift outside its support, show that there is no contribution outside this degree sector. Both radial boundary coefficients in (4.5) are zero.

For an explicit self-adjointness check, the norm formula (4.3) shows that the radial inner product, up to a fixed positive multiplier, has weights

$$\nu_\ell(k) = \frac{(\beta n)^k}{(k-1)!}\binom{n-2\ell}{k-\ell}. \tag{4.7}$$

They satisfy

$$\nu_\ell(k)\,\beta(n-k-\ell) = \nu_\ell(k+1)\,\frac{k(k+1-\ell)}{n}.$$

The killing term is nonnegative. If the sector has more than one layer, its interior birth and death rates are positive. Choosing an orthogonal basis in each H_ℓ therefore yields orthogonal radial copies of (4.5). Completeness follows from Lemma 4.1, proving the spectral assertion. □

Proposition 4.3. *Every rotation-invariant function F on Ω_n satisfies*

$$\mathcal{E}_B(F,F) \geq \left(\beta + \frac{2}{n}\right)\mathrm{Var}_{\pi_B}(F). \tag{4.8}$$

In particular,

$$\gamma_M \geq \beta + \frac{2}{n}. \tag{4.9}$$

Proof. For $\ell\geq 2$, the killing in (4.5) is at least

$$\ell\beta + \frac{\ell(\ell-1)}{n} \geq 2\beta + \frac{2}{n}.$$

By Lemma 3.1 with $h=1$, every eigenvalue of $-\mathcal{L}_\ell$ is at least this value. In degree zero, every nonconstant eigenvalue is at least $\beta+2/n$ by (3.6).

It remains to exclude degree one. The degree-one part on any layer $1\leq k\leq n-1$ has the unique form

$$S \longmapsto \sum_{i\in S} a_i(k), \qquad \sum_{i=1}^n a_i(k) = 0.$$

Uniqueness follows from the injectivity in Lemma 4.1. All degree sectors and their orthogonal complements are preserved by relabelling. Since the inner product is also preserved, their orthogonal projections commute with rotations. The degree-one projection of a rotation-invariant F is therefore rotation-invariant. Its unique coefficient vector must be fixed by the cyclic permutation of all n labels, so its entries are all equal. Their sum is zero, and the vector vanishes. The lift is already zero on layer n.

After subtracting its mean, F consequently contains only the nonconstant part of degree zero and degrees at least two. The complete orthogonal spectral decomposition now proves (4.8). Apply Proposition 2.1 to conclude (4.9). □

5. A reciprocal certificate in the square-root regime

Proposition 5.1. *If $n \ge 8$ and $4/n^2 \le \beta \le 1/16$, then*

$$\gamma_B \ge \frac{\sqrt{\beta}}{5}, \qquad \gamma_M \ge \frac{\sqrt{\beta}}{5}. \tag{5.1}$$

Proof. Put $a = n\sqrt{\beta}$, so that $2 \le a \le n/4$. Use the positive test vector $h(k) = 1/(k+a)$, first for the count derivative $\mathcal{A}$. Its exact quotient is

$$\begin{aligned} \frac{-(\mathcal{A}h)(k)}{h(k)} &= \frac{\beta(n-k-1)}{k+a+1} - \frac{k(k-1)}{n(k+a-1)} + \beta + \frac{2k}{n} \\ &\ge \frac{\beta(n-k-1)}{k+a+1} + \beta + \frac{k}{n}. \end{aligned} \tag{5.2}$$

If $k \le a$, then $n-k-1 \ge n/2$ and $k+a+1 \le 5a/2$. The birth term in the last line is therefore at least $\beta n/(5a) = \sqrt{\beta}/5$. If $k \ge a$, its last term is at least $a/n = \sqrt{\beta}$. Lemma 3.1 proves that every nonconstant degree-zero eigenvalue is at least $\sqrt{\beta}/5$.

For degree $\ell \ge 1$, the same test vector on $\ell \le k \le n-\ell$ gives

$$\frac{-(\mathcal{L}_\ell h)(k)}{h(k)} = \frac{\beta(n-k-\ell)}{k+a+1} + \ell\beta + \frac{(k-1)((\ell-1)k+\ell a)}{n(k+a-1)}. \tag{5.3}$$

For clarity, the last term combines the negative death contribution $-(k-1)(k-\ell)/(n(k+a-1))$ with the killing $\ell(k-1)/n$. Their numerator is

$$(k-1)(\ell(k+a-1) - (k-\ell)) = (k-1)((\ell-1)k+\ell a).$$

All terms on the right of (5.3) are nonnegative. If $k \le a$, then $\ell \le k$ gives $n-k-\ell \ge n-2a \ge n/2$, while $k+a+1 \le 5a/2$. The birth term is again at least $\sqrt{\beta}/5$. If $k \ge a$, then $(\ell-1)k+\ell a \ge a$ and

$$\frac{k-1}{k+a-1} \ge \frac{1}{3}, \qquad 3(k-1) - (k+a-1) = 2k-a-2 \ge a-2 \ge 0.$$

The last term in (5.3) is at least $a/(3n) = \sqrt{\beta}/3$. Another application of Lemma 3.1 bounds every eigenvalue of every $-\mathcal{L}_\ell$ below by $\sqrt{\beta}/5$.

Lemma 4.2 exhausts the spectrum of the cut chain, so $\gamma_B \ge \sqrt{\beta}/5$. The polymer inequality follows from Proposition 2.1. □

6. Completion of the uniform bound

Proof of Theorem 1.1. Write $s = \beta + \sqrt{\beta} + 1/n$. Proposition 3.2 gives $\gamma_M \le 4s$. For the lower bound, use (4.9) throughout and divide the parameters into the following exhaustive cases.

If $\beta \ge 1/16$, then $\sqrt{\beta} \le 4\beta$, whence

$$s \le 5\beta + \frac{1}{n} \le 5\left(\beta + \frac{2}{n}\right) \le 5\gamma_M.$$

If $\beta \le 4/n^2$, then $\sqrt{\beta} \le 2/n$, so

$$s \le \beta + \frac{3}{n} \le \frac{3}{2}\left(\beta + \frac{2}{n}\right) \le \frac{3}{2}\gamma_M.$$

If $n < 8$ and $\beta < 1/16$, the same conclusion follows from $\sqrt{\beta} < 1/4 \le 2/n$.

In every remaining case, $n \ge 8$ and $4/n^2 < \beta < 1/16$. Proposition 5.1 gives $\sqrt{\beta} \le 5\gamma_M$, while (4.9) gives $\beta + 1/n \le \gamma_M$. Adding these inequalities yields $s \le 6\gamma_M$. The preceding weak inequalities

cover all boundary values. Therefore $s/6 \le \gamma_M \le 4s$ for every $n \ge 2$ and $\beta > 0$. Both bounds are positive, so their reciprocals give the stated relaxation-time bounds. □

7. Exact reduction and sharp spectral asymptotics

The preceding argument controls every polymer observable without assuming that the full gap is attained by a count observable. We now prove that it is attained there. Granovsky and Erlihson [15, Section 2.3] announce this equality for their solvable class. The proof below controls every orthogonal sector explicitly, and the same comparison is used for the spatial extension in Section 11.

Lemma 7.1 (Adjacent ratios). *Let $h > 0$ be a ground state of the count-derivative operator $\mathcal{A}$ in (3.5): $-\mathcal{A}h = \gamma_Q h$. Then*

$$\frac{h(i+1)}{h(i)} > \frac{1}{2}, \qquad 1 \le i \le n-2.$$

For $n = 2$ the assertion has no ratios to check.

Proof. Write $\lambda_i = \beta(n-i-1)$, $\mu_i = i(i-1)/n$ and $V_i = \beta + 2i/n$. Suppose that $r = h(i+1)/h(i) \le 1/2$ is the minimum of the forward ratios. The eigenvalue equations at i and $i+1$, respectively, imply

$$\gamma_Q \ge \lambda_i(1-r) + \mu_i(1-r^{-1}) + V_i,$$
$$\gamma_Q \le \lambda_{i+1}(1-r) + \mu_{i+1}(1-r^{-1}) + V_{i+1}.$$

Indeed, the preceding ratio is at least r in the first inequality, and the following ratio is at least r in the second. A missing preceding or following ratio is multiplied by a zero boundary rate. Subtracting gives

$$0 \ge \beta(1-r) + \frac{2i}{n}(r^{-1}-1) - \frac{2}{n} > 0.$$

At $r = 1/2$ the last expression equals $\beta/2 + 2(i-1)/n > 0$; it increases as r decreases. This contradiction proves the assertion. □

Proposition 7.2 (Comparison of the higher sectors). *For $2 \le \ell \le \lfloor n/2 \rfloor$,*

$$\inf \mathrm{Spec}(-\mathcal{L}_\ell) \ge \gamma_Q + \frac{(\ell-2)(\ell+1)}{n}.$$

In fact every sector with $\ell \ge 2$ has its bottom eigenvalue strictly above γ_Q.

Proof. Set $r = k - \ell + 1$ and use $g(k) = h(r)$ as a positive test vector for (4.5). Its domain is $1 \le r \le n - 2\ell + 1$. Comparing the birth, death and killing coefficients with those of $\mathcal{A}$ gives

$$-\frac{\mathcal{L}_\ell g(k)}{g(k)} = \gamma_Q + (\ell-1)\beta\left(2\frac{h(r+1)}{h(r)} - 1\right) + \frac{\ell-2}{n}\left[r + \ell - (r-1)\left(\frac{h(r-1)}{h(r)} - 1\right)\right]. \tag{7.1}$$

The forward-ratio term is strictly positive by Lemma 7.1. The backward ratio is at most two, so the expression in square brackets is at least $\ell + 1$. The positive-vector comparison proves the claim. At $r = 1$ the backward term is zero. At the upper endpoint, $r + 1 = n - 2\ell + 2 \le n - 1$, so the forward value is in the domain of h. The same computation applies to a one-state sector. □

Theorem 7.3 (Exact count reduction). *For every $n \ge 2$ and $\beta > 0$,*

$$\boxed{\gamma_M(n,\beta) = \gamma_Q(n,\beta).}$$

More generally, the best Poincaré constant on rotation-invariant cut-set observables is γ_Q.

Proof. Rotation invariance eliminates degree one, as proved in Proposition 4.3. The nonconstant degree-zero sector has lowest rate γ_Q, and Proposition 7.2 puts every surviving higher sector at or above that rate. Orthogonality therefore gives a Poincaré inequality with constant γ_Q on all rotation-invariant functions. The stationary comparison transfers it to polymers. The opposite inequality follows by testing with count observables. For $n = 2, 3$ there is no higher sector to check. □

7.1. **An explicit error bound for the leading constant.** The same family of positive vectors used for the earlier upper bound gives a sharper result if its parameter is chosen by an algebraic cancellation.

Theorem 7.4 (Sharp estimate with a uniform error)**.** *Put*

$$d = \sqrt{\beta^2 + 4\beta}, \qquad a = \frac{\beta n + 1 + \sqrt{(\beta n + 1)^2 + 4\beta n(n+1)}}{2}, \qquad c = \frac{2a}{n} - \beta.$$

Then $a > 1$ and

$$c - \frac{6(a-1)}{n(a+3)} \le \gamma_M \le c - \frac{6a(a-1)}{n(n+a-2)(n+a+1)}. \tag{7.2}$$

In particular,

$$\boxed{d - \frac{5}{n} \le \gamma_M(n, \beta) \le d + \frac{4}{n}.} \tag{7.3}$$

These estimates can be combined with $\gamma_M \ge \beta + 2/n$ whenever the lower expression is small.

Proof. Choose $h(k) = [(k+a)(k+a+1)]^{-1}$ for the count derivative. Its forward and backward ratios are respectively $1 - 2/(k+a+2)$ and $1 + 2/(k+a-1)$. Hence

$$\begin{aligned} -\frac{\mathcal{A}h(k)}{h(k)} &= \beta + \frac{2\beta(n-k-1)}{k+a+2} + \frac{2ak}{n(k+a-1)} \\ &= \frac{2a}{n} - \beta + \frac{2\beta(n+a+1)}{k+a+2} - \frac{2a(a-1)}{n(k+a-1)}. \end{aligned}$$

The definition of a is precisely $a(a-1) = \beta n(n+a+1)$. Thus

$$-\frac{\mathcal{A}h(k)}{h(k)} = c - \frac{6a(a-1)}{n(k+a-1)(k+a+2)}. \tag{7.4}$$

This quantity increases with k; its extreme values at 1 and $n-1$ give (7.2) by positive-vector comparison and Theorem 7.3. Both endpoint terms in the original generator are treated with their zero coefficients, so the calculation also holds for $n = 2$.

The subtracted term in (7.4) is between zero and $6/n$. Moreover,

$$c = \frac{1}{n} + \sqrt{d^2 + \frac{6\beta}{n} + \frac{1}{n^2}}, \qquad d + \frac{1}{n} \le c \le d + \frac{4}{n}.$$

For the upper inequality, square $d + 3/n$ and use $d \ge \beta$. The lower inequality is immediate. These facts imply (7.3). □

Corollary 7.5 (Asymptotic equivalents)**.** *Let $n \to \infty$ and allow $\beta = \beta_n > 0$ to vary. If $n\sqrt{\beta_n} \to \infty$, then*

$$\gamma_M(n, \beta_n) \sim \sqrt{\beta_n^2 + 4\beta_n}.$$

In particular,

$$\begin{aligned} \beta_n \sim b/n,\ b > 0 \quad &\Longrightarrow \quad \sqrt{n}\,\gamma_M(n, \beta_n) \longrightarrow 2\sqrt{b}, \\ \beta_n \longrightarrow \beta_0 > 0 \quad &\Longrightarrow \quad \gamma_M(n, \beta_n) \longrightarrow \sqrt{\beta_0^2 + 4\beta_0}. \end{aligned}$$

If $n^2\beta_n \to 0$, then

$$n\gamma_M(n,\beta_n) \longrightarrow 2.$$

Proof. In the first regime, $nd \geq 2n\sqrt{\beta_n} \to \infty$, so the additive error in (7.3) is negligible. The two displayed specializations follow. In the final regime, $a \to 1$ in Theorem 7.4; its upper bound gives $n\gamma_M \leq 2a - \beta_n n \to 2$, while the lower bound $\gamma_M \geq \beta_n + 2/n$ gives the reverse limit. □

8. The critical scaling function

The transition scale $\beta \asymp n^{-2}$ admits a limiting spectral problem with an explicit finite-size error. We write $\gamma_Q(n,\beta)$ for the gap of the count chain with birth rates $\beta(n-k)$ and death rates $k(k-1)/n$. By Theorem 7.3, it equals the polymer gap $\gamma_M(n,\beta)$.

For $b > 0$, define a killed birth–death operator on the positive integers by

$$(\mathcal{A}_b f)(k) = b[f(k+1) - f(k)] + k(k-1)[f(k-1) - f(k)] - 2kf(k), \qquad k \geq 1. \tag{8.1}$$

The backward term at $k = 1$ is zero. Its underlying conservative chain has reversible probability measure

$$\pi_b(k) = Z_b^{-1}\frac{b^{k-1}}{k!(k-1)!}, \qquad Z_b = \sum_{k\geq 1}\frac{b^{k-1}}{k!(k-1)!}. \tag{8.2}$$

All moments of π_b are finite.

Proposition 8.1 (Construction of the limiting operator)**.** *On $\ell^2(\pi_b)$, the quadratic form*

$$\mathcal{E}_b(f,f) = \sum_{k\geq 1}\pi_b(k)b[f(k+1) - f(k)]^2 + \sum_{k\geq 1} 2k\pi_b(k)f(k)^2 \tag{8.3}$$

with domain consisting of the functions for which this expression is finite is densely defined and closed. Its associated self-adjoint operator $H_b = -\mathcal{A}_b$ has compact resolvent. Its smallest eigenvalue $\rho(b)$ is simple and has a strictly positive eigenfunction h_b.

Proof. Detailed balance follows from $\pi_b(k)b = \pi_b(k+1)k(k+1)$. Closedness of the form follows by taking coordinate limits of a Cauchy sequence in the form norm and using the nonnegative sums in (8.3). Finitely supported functions form a core. Indeed, for f in the form domain, truncate it at R. The only extra edge term in the squared form norm of the discarded tail is $\pi_b(R)bf(R+1)^2$, and

$$\pi_b(R)bf(R+1)^2 \leq 2\pi_b(R)b[f(R+1) - f(R)]^2 + 2\pi_b(R)bf(R)^2 \longrightarrow 0.$$

All the other terms are tails of convergent nonnegative series.

The form domain is compactly embedded in $\ell^2(\pi_b)$: for every $R \geq 1$,

$$\sum_{k>R}\pi_b(k)f(k)^2 \leq \frac{\mathcal{E}_b(f,f)}{2(R+1)}. \tag{8.4}$$

Thus a bounded set in the form norm has uniformly small tails, while each finite-coordinate projection is precompact. The associated operator therefore has compact resolvent, and the bottom of its spectrum is attained. Taking the absolute value does not increase the form. A nonnegative eigenfunction at the bottom is strictly positive: the coordinate eigenvalue equation propagates any zero to both neighbors, and hence to every state. Finally, the equation at $k = 1$ determines $h(2)$ from $h(1)$; the subsequent three-term recurrence determines all remaining coordinates. The lowest eigenspace is therefore one-dimensional. □

The next ratio estimate will control both the limit and its finite approximations. The finite count-derivative operator is

$$\begin{aligned}(A_{n,\beta}h)(k) &= \beta(n-k-1)[h(k+1)-h(k)] \\ &\quad + \frac{k(k-1)}{n}[h(k-1)-h(k)] - \left(\beta + \frac{2k}{n}\right)h(k), \qquad 1 \le k \le n-1.\end{aligned} \tag{8.5}$$

Its spectrum is the nonzero spectrum of the count generator, with the same signs.

Lemma 8.2 (Ground-state ratios)**.** *Let h be a positive ground state of either $-A_{n,\beta}$ or H_b. At every pair of successive states in its domain,*

$$\frac{k+1}{k+3} \le \frac{h(k+1)}{h(k)} \le 1. \tag{8.6}$$

In particular, h_b is bounded.

Proof. A nonexplosive birth–death chain on an interval has an order-preserving coupling. Run two copies independently until they meet and then coalesce them. Nearest-neighbor paths cannot cross without meeting. Consequently the Feynman–Kac survival function

$$k \longmapsto \mathbb{E}_k \exp\left\{-\int_0^t V(X_s)\,ds\right\}$$

is decreasing when the killing potential V is increasing, and is increasing when V is decreasing. The conservative chain underlying $\mathcal{A}_b$ is nonexplosive: its births have constant rate b, and the number of deaths by any time is at most the initial population minus one plus the number of births. All finite chains here are nonexplosive as well.

For the original killed operators the potentials are $\beta + 2k/n$ and $2k$, respectively, and hence are increasing. Their survival functions are decreasing in k. The spectral projection onto the simple positive ground state, after multiplication by the exponential of the lowest eigenvalue, converges in the reversible ℓ^2 space and therefore at each coordinate. Applied to the constant function, this shows that h is decreasing.

For the other inequality set

$$h_0(k) = \frac{1}{(k+1)(k+2)}.$$

Conjugating the killed operator by multiplication by h_0 gives a birth–death operator whose new killing potential is, respectively,

$$\begin{aligned}-\frac{A_{n,\beta}h_0(k)}{h_0(k)} &= \beta + \frac{2}{n} + \frac{2\beta(n-k-1)}{k+3} = \frac{2}{n} - \beta + \frac{2\beta(n+2)}{k+3}, \\ -\frac{\mathcal{A}_b h_0(k)}{h_0(k)} &= 2 + \frac{2b}{k+3}.\end{aligned} \tag{8.7}$$

Both are positive and decreasing. The transformed birth and death rates in the infinite case are

$$b\frac{k+1}{k+3}, \qquad (k-1)(k+2).$$

Its birth rates are bounded by b, so the same argument proves nonexplosion. The transformed reversible measure is proportional to $\pi_b h_0^2$, and multiplication by h_0 is a unitary identification with the original weighted space, up to an irrelevant normalization. The transformed ground state is h/h_0. Applying the preceding survival-function argument to the decreasing potential shows that h/h_0 is increasing. Since $h_0(k+1)/h_0(k) = (k+1)/(k+3)$, this proves the lower ratio bound.

For completeness, in the infinite case the Feynman–Kac semigroups used here are the semigroups of the form operators. One may first kill the conservative paths upon leaving a finite interval and then let that interval increase. Nonexplosion gives the pathwise limit, and the finitely supported form core established above identifies its operator with H_b. The same argument applies after

conjugation by h_0. Thus the spectral projection and the order comparison concern the same semigroup. □

Theorem 8.3 (Critical limit with a finite-size error)**.** *For all $n \geq 2$ and $b > 0$,*

$$\left| n\gamma_M\left(n, \frac{b}{n^2}\right) - \rho(b) \right| \leq \frac{b}{n}. \tag{8.8}$$

The same statement holds with γ_Q in place of γ_M.

Proof. Restrict h_b to $1, \ldots, n-1$. The infinite eigenvalue equation gives the exact identity

$$-\frac{nA_{n,b/n^2}h_b(k)}{h_b(k)} = \rho(b) + \frac{b}{n}\left[1 + (k+1)\left(\frac{h_b(k+1)}{h_b(k)} - 1\right)\right]. \tag{8.9}$$

This includes the upper endpoint $k = n-1$: the finite birth rate there is zero, and the displayed correction accounts for removal of the infinite birth term. By (8.6), the bracket lies between $-1+4/(k+3)$ and 1, and hence between -1 and 1. Positive-vector comparison for the finite killed chain proves the claim for γ_Q. Theorem 7.3 then gives the claim for γ_M. □

Proposition 8.4 (Continuity and moving critical parameters)**.** *For $0 < b_1 \leq b_2$,*

$$0 \leq \rho(b_2) - \rho(b_1) \leq \frac{b_2 - b_1}{2}. \tag{8.10}$$

Moreover,

$$2 < \rho(b) \leq 2 + \frac{b}{2} \qquad (b > 0). \tag{8.11}$$

Thus ρ extends continuously to $b = 0$ by $\rho(0) = 2$. If $\beta_n > 0$ and $b_n = n^2\beta_n \longrightarrow b \in [0, \infty)$, then

$$|n\gamma_M(n, \beta_n) - \rho(b)| \leq \frac{b_n}{n} + \frac{|b_n - b|}{2} \longrightarrow 0. \tag{8.12}$$

Proof. Write $h_i = h_{b_i}$ and $\delta = b_2 - b_1$. Pointwise,

$$H_{b_2}h_1 = \rho(b_1)h_1 + \delta[h_1(k) - h_1(k+1)].$$

The ratio lemma implies

$$0 \leq \frac{h_1(k) - h_1(k+1)}{h_1(k)} \leq \frac{2}{k+3} \leq \frac{1}{2}. \tag{8.13}$$

This identity may legitimately be paired with h_2 in $\ell^2(\pi_{b_2})$. Indeed, h_1 is bounded; the weight π_{b_2} has all moments, so h_1 is in the form domain of H_{b_2}. The displayed pointwise expression for $H_{b_2}h_1$ is bounded and hence square integrable. Summation by parts against finitely supported test functions, followed by the form-core approximation, puts h_1 in the operator domain. Self-adjointness therefore yields

$$[\rho(b_2) - \rho(b_1)]\langle h_2, h_1\rangle_{\pi_{b_2}} = \delta\langle h_2, h_1 - h_1(\cdot + 1)\rangle_{\pi_{b_2}}.$$

The inner product on the left is strictly positive. Dividing and using (8.13) proves (8.10).

The potential term in (8.3) gives $\rho(b) \geq 2$; strict positivity of the ground state at states $k > 1$ makes the inequality strict. At $k = 1$ the eigenvalue equation reads

$$\rho(b) = 2 + b\left(1 - \frac{h_b(2)}{h_b(1)}\right) \leq 2 + \frac{b}{2}.$$

This proves the extension at zero. Finally, (8.12) follows from Theorem 8.3 and the Lipschitz estimate. □

Proposition 8.5 (Finite-matrix brackets)**.** *For an integer $R \geq 1$, suppress the birth at R in (8.1) and retain the same killing $2k$ on $1, \dots, R$. Let $\rho_R(b)$ be the smallest eigenvalue of the negative of this finite killed generator. Then*

$$\rho_R(b) \leq \rho(b) \leq \rho_R(b) + \frac{2b}{R+3}. \tag{8.14}$$

Equivalently, $\rho_R(b)$ is the smallest eigenvalue of the symmetric tridiagonal matrix $J_R(b)$ with entries

$$\begin{aligned} (J_R)_{kk} &= k(k+1) + b\,\mathbf{1}_{\{k<R\}}, \qquad 1 \leq k \leq R, \\ (J_R)_{k,k+1} &= (J_R)_{k+1,k} = -\sqrt{bk(k+1)}, \qquad 1 \leq k < R. \end{aligned} \tag{8.15}$$

Proof. Restrict h_b to $1, \dots, R$. Its positive-vector quotient is $\rho(b)$ at all states below R, while at R it is

$$\rho(b) - b\left(1 - \frac{h_b(R+1)}{h_b(R)}\right) \in \left[\rho(b) - \frac{2b}{R+3}, \rho(b)\right].$$

Finite positive-vector comparison proves (8.14), including $R = 1$. Conjugation by the square root of the finite reversible weights gives (8.15). □

The brackets above give an explicit approximation of the critical scaling function without an unquantified infinite-state truncation. If a finite-matrix computation supplies rigorous bounds $a_R \leq \rho_R(b) \leq c_R$, then $[a_R, c_R + 2b/(R+3)]$ is a rigorous interval for $\rho(b)$.

8.1. **The degree-one critical scaling function.** The degree-one eigenvalue in the spatial theorem has a parallel limit. On the nonnegative integers define

$$(\mathcal{A}_b^{(1)} f)(j) = b[f(j+1) - f(j)] + j^2[f(j-1) - f(j)] - jf(j), \qquad j \geq 0. \tag{8.16}$$

Its reversible weights are

$$\nu_b(j) = \widehat{Z}_b^{-1} \frac{b^j}{(j!)^2}.$$

Use the closed form

$$\mathcal{F}_b(f,f) = \sum_{j \geq 0} \nu_b(j) b[f(j+1) - f(j)]^2 + \sum_{j \geq 0} j\nu_b(j) f(j)^2$$

to define $H_b^{(1)} = -\mathcal{A}_b^{(1)}$. The form-core and compactness arguments of Proposition 8.1 apply verbatim, with the tail bound $\sum_{j>R} \nu_b(j) f(j)^2 \leq \mathcal{F}_b(f,f)/(R+1)$. Thus $H_b^{(1)}$ has a simple positive ground state g_b and a smallest eigenvalue $\sigma(b) > 0$. Positivity of this eigenvalue follows because zero form energy would force a constant function vanishing at every $j \geq 1$.

Proposition 8.6 (Degree-one critical approximation)**.** *For all $n \geq 2$ and $b > 0$,*

$$0 \leq n\eta_1\left(n, \frac{b}{n^2}\right) - \sigma(b) \leq \frac{b}{n}. \tag{8.17}$$

The limiting eigenvalue satisfies

$$\frac{b}{\sqrt{b}+2} \leq \sigma(b) \leq \min\left\{\sqrt{b}, \frac{b}{2}\right\}. \tag{8.18}$$

It is increasing and $1/2$-Lipschitz, and extends continuously to zero by $\sigma(0) = 0$.

Proof. The killing j is increasing, so the order-preserving survival comparison used in Lemma 8.2 makes g_b decreasing. Conjugation by $g_0(j) = 1/(j+1)$ yields the decreasing killing potential $b/(j+2)$ and birth and death rates $b(j+1)/(j+2)$ and $j(j+1)$. Its births are bounded and its chain is nonexplosive. The same transformed-semigroup argument gives

$$\frac{j+1}{j+2} \leq \frac{g_b(j+1)}{g_b(j)} \leq 1. \tag{8.19}$$

Set $j = k - 1$ in the finite operator (11.3) and test it with g_b. Its exact quotient is

$$-\frac{nA_1 g_b(j)}{g_b(j)} = \sigma(b) + \frac{b}{n}\left[1 + (j+2)\left(\frac{g_b(j+1)}{g_b(j)} - 1\right)\right], \qquad 0 \le j \le n-2,$$

where A_1 here is expressed in the shifted coordinate. The bracket belongs to $[0, 1]$, including the upper endpoint where the finite birth rate vanishes. Positive-vector comparison proves (8.17).

For the bounds put $a = \sqrt{b}$ and test the infinite operator with $f(j) = 1/(j+1+a)$. Its quotient is

$$-\frac{\mathcal{A}_b^{(1)} f(j)}{f(j)} = a - \frac{2a^2}{(j+a)(j+a+2)} \in \left[\frac{a^2}{a+2}, a\right].$$

This test function is bounded and belongs to the operator domain; pairing its equation with the positive ground state justifies the infinite-state comparison, exactly as in the proof of Proposition 8.4. The equation at $j = 0$ also gives $\sigma(b) = b[1 - g_b(1)/g_b(0)] \le b/2$. Finally, changing b_1 to $b_2 \ge b_1$ adds $(b_2 - b_1)[1 - g_{b_1}(j+1)/g_{b_1}(j)]$ to the quotient. This lies in $[0, (b_2 - b_1)/2]$ by (8.19). The cross-parameter domain argument and positive-ground-state pairing in Proposition 8.4 apply with ν_b in place of π_b. They prove the stated monotonicity and Lipschitz bound; the estimate $\sigma(b) \le b/2$ supplies continuity at zero. □

Corollary 8.7 (The ends of the crossover)**.** *The limiting functions satisfy*

$$\begin{aligned} &\rho(b) \longrightarrow 2, \qquad \sigma(b) \sim b/2 \quad \text{as } b \downarrow 0, \\ &\rho(b) \sim 2\sqrt{b}, \qquad \sigma(b) \sim \sqrt{b} \quad \text{as } b \longrightarrow \infty. \end{aligned} \tag{8.20}$$

In particular, the count and spatial degree-one modes have different leading constants at the large-b end of the critical scaling.

Proof. The assertions for σ follow by squeezing with (8.18); the limit of ρ at zero was proved in Proposition 8.4. For fixed $b > 0$, apply the additive-error estimate of Theorem 7.4 at $\beta = b/n^2$, multiply by n, and then let $n \to \infty$ using Theorem 8.3. This gives

$$2\sqrt{b} - 5 \le \rho(b) \le 2\sqrt{b} + 4,$$

which proves the last asymptotic equivalent. □

For numerical approximation of σ, suppress the birth at an integer $R \ge 0$ and retain the killing j on $0, \dots, R$. Let $\sigma_R(b)$ be the lowest eigenvalue of this finite operator. The same restricted-ground-state argument as in Proposition 8.5, now using (8.19), gives

$$0 \le \sigma(b) - \sigma_R(b) \le \frac{b}{R+2}. \tag{8.21}$$

Indeed, only the last quotient changes, by $-b[1 - g_b(R+1)/g_b(R)]$. The corresponding symmetric matrix has diagonal $j(j+1) + b\mathbf{1}_{\{j<R\}}$ and off-diagonal $-\sqrt{b}(j+1)$ between j and $j+1$.

Corollary 8.8 (The joint critical spatial limit)**.** *For every $n \ge 2$, $b > 0$, connected weighted graph G on n vertices and $D \ge 0$,*

$$\left| n\gamma_{\mathrm{sp}}\left(n, \frac{b}{n^2}, D, G\right) - \min\{\rho(b), \sigma(b) + nD\lambda_G\} \right| \le \frac{b}{n}. \tag{8.22}$$

Consequently, if $n^2\beta_n \to b \in [0, \infty)$ and $nD_n\lambda_{G_n} \to \delta \in [0, \infty)$, then

$$n\gamma_{\mathrm{sp}}(n, \beta_n, D_n, G_n) \longrightarrow \min\{\rho(b), \sigma(b) + \delta\}. \tag{8.23}$$

If instead $nD_n\lambda_{G_n} \to \infty$, the limit is $\rho(b)$.

Proof. Apply Theorem 11.1, Theorem 8.3, and Proposition 8.6. The minimum of two real numbers is 1-Lipschitz for their maximum-coordinate distance. For moving parameters use the continuity of ρ and σ. □

Figure 1 illustrates the inverse-mass limit and the two critical scaling functions.

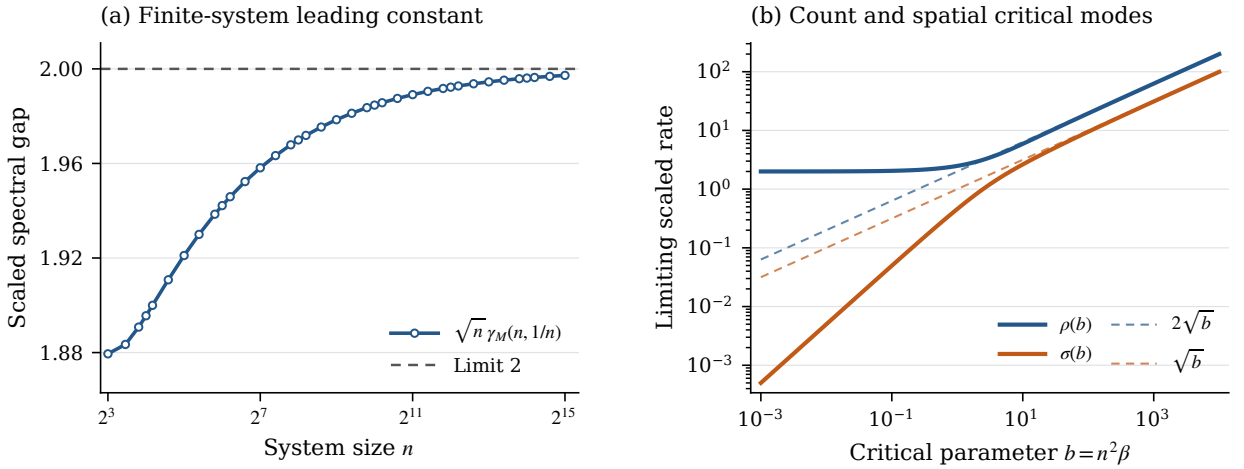


FIGURE 1. Spectral scaling from finite radial matrices. (a) Direct numerical eigenvalues of the count derivative give $\sqrt{n}\,\gamma_M(n,1/n)$ approaching 2. (b) Finite Jacobi approximations to $\rho(b)$ and $\sigma(b)$, with their large-b equivalents $2\sqrt{b}$ and $\sqrt{b}$. The faint bands show the analytic truncation widths $2b/(R+3)$ and $b/(R+2)$, respectively, where $R=\lceil 600(1+\sqrt{b})\rceil$. Finite eigenvalues are evaluated in floating point; the bands do not include a certified rounding-error interval. Source data and the plotting program accompany the paper.

9. TOTAL-VARIATION AND SEPARATION CUTOFF

For a finite chain with semigroup P_t and stationary law π, write

$$d_x(t)=\|P_t(x,\cdot)-\pi\|_{\mathrm{TV}},\qquad d(t)=\max_x d_x(t),\qquad t_{\mathrm{mix}}(\varepsilon)=\inf\{t:d(t)\le\varepsilon\}.$$

The superscripts "count" and "polymer" specify the chain. For a specified initial state the maximum is omitted. All logarithms are natural. The time-window statements below imply cutoff because the window is smaller than the stated mixing time by a factor tending to zero.

9.1. **The two polymer endpoints.** Let $L=L_M$ be the polymer generator, with stationary law $\pi=\pi_M$, and write $K(x)$ for the number of polymers. The count strongly lumps L to the birth–death generator Q on $\{1,\dots,n\}$. Let $\bar\pi$ denote the pushforward of π by K. For every function g on $\{1,\dots,n\}$,

$$L(g\circ K)=(Qg)\circ K,\qquad P_t(g\circ K)=(\bar P_t g)\circ K.$$

Fix i such that the fiber $K^{-1}(i)$ consists of one state x_i. The initial density with respect to stationarity is then

$$h_0(x)=\frac{\mathbf{1}_{\{x_i\}}(x)}{\pi(x_i)}=\frac{\mathbf{1}_{\{i\}}(K(x))}{\bar\pi(i)}.$$

By reversibility, the density at time t is $P_t h_0$; invariance of the count-function subspace gives

$$\frac{P_t(x_i,x)}{\pi(x)}=\frac{\bar P_t(i,K(x))}{\bar\pi(K(x))}.$$

Consequently, at every $t\ge 0$,

$$\boxed{\|P_t(x_i,\cdot)-\pi\|_{\mathrm{TV}}=\|\bar P_t(i,\cdot)-\bar\pi\|_{\mathrm{TV}}.}$$

Indeed, sum the absolute density difference over each count fiber and use $\sum_{K(x)=k}\pi(x)=\bar{\pi}(k)$. Both $i=1$ (one polymer of size n) and $i=n$ (all monomers) have singleton fibers, so the identity holds at both endpoints of the polymer chain. It transfers endpoint cutoff, and provides a lower bound on worst-case polymer mixing. It does not provide an upper bound for arbitrary initial polymer states.

9.2. **Count cutoff for fixed scaled fragmentation rate.** Throughout this section, $b>0$ is fixed and constants may depend on b. Let Q_n be the continuous-time birth–death generator on $\{1,\dots,n\}$ with rates

$$\lambda_k=\frac{b(n-k)}{n},\qquad \mu_k=\frac{k(k-1)}{n}.$$

Its stationary law $p=p_n$ satisfies

$$p(k)\propto\binom{n}{k}\frac{b^k}{(k-1)!},\qquad \frac{p(k+1)}{p(k)}=\frac{b(n-k)}{k(k+1)}.$$

Define the positive drift zero

$$a=a_n=\frac{-(b-1)+\sqrt{(b-1)^2+4bn}}{2}=\sqrt{bn}+O_b(1),$$

so that

$$D_k:=\mu_k-\lambda_k=\frac{(k-a)(k+a+b-1)}{n}.$$

Theorem 9.1 (Count cutoff and polymer endpoint cutoff)**.** *For every fixed $b>0$, the count chains have a worst-case total-variation cutoff at*

$$t_n=\frac{\sqrt{n}\log n}{8\sqrt{b}}$$

with an $O_b(\sqrt{n})$ window. Precisely, for each $\varepsilon\in(0,1)$,

$$t^{\mathrm{count}}_{\mathrm{mix},n}(\varepsilon)=t_n+O_{b,\varepsilon}(\sqrt{n}).$$

The same statement holds separately for the count chain started from 1 and from n. The polymer chain at $\beta=b/n$ strongly lumps to Q_n. Its total-variation distance from the single n-polymer and all-monomer states equals the corresponding count distance at every time. These two polymer initial states therefore have the same cutoff and window. In particular, the worst-case polymer mixing time is at least $t_n-O_{b,\varepsilon}(\sqrt{n})$. An arbitrary-start upper bound is proved in Theorem 10.1.

9.3. **Elementary stationary estimates.** Write $F_k=p([1,k])$ and $G_k=p([k,n])$. We shall use the following bounds:

$$\mathbb{E}_p|K-a|\le C_b\sqrt{a}, \tag{9.1}$$

$$c_{b,L}a^{-1/2}\le p(k)\le C_{b,L}a^{-1/2}\qquad(|k-a|\le L\sqrt{a}), \tag{9.2}$$

$$F_k\wedge G_k\ge c_{b,L}\qquad(|k-a|\le L\sqrt{a}). \tag{9.3}$$

Here L is any fixed finite constant and n is sufficiently large. In particular every stationary median m satisfies $|m-a|\le C_b\sqrt{a}$.

For completeness, these estimates follow directly from the consecutive weight ratio. For unnormalized stationary weights w_k, set

$$R(x)=\frac{b(n-x)}{x(x+1)}.$$

It is decreasing, and its unique positive solution to $R(x) = 1$ is $a + O_b(1)$. A maximizing index r for w_k therefore satisfies $r = a + O_b(1)$. On $[a/2, 2a]$, for all sufficiently large n,

$$-\frac{C_b}{a} \le (\log R)'(x) = -\frac{1}{n-x} - \frac{1}{x} - \frac{1}{x+1} \le -\frac{c_b}{a}.$$

Since $R(r-1) \ge 1 \ge R(r)$, summing these derivative bounds over successive ratios gives

$$c_b \exp\left(-C_b \frac{(k-a)^2}{a}\right) \le \frac{w_k}{w_r} \le C_b \exp\left(-c_b \frac{(k-a)^2}{a}\right), \qquad a/2 \le k \le 2a.$$

Rounding endpoints changes only the constants. Below $a/2$, the successive ratios in the direction away from the mode are bounded by a fixed number $q_b < 1$; the same holds above $2a$. These two tails are thus geometric, starting at weights at most $C_b e^{-c_b a} w_r$. Consequently

$$\sum_k w_k \asymp_b \sqrt{a}\, w_r, \qquad \sum_k |k-a| w_k \le C_b a\, w_r.$$

This proves (9.1) and (9.2). To obtain (9.3), sum the local lower bound over an interval of length $\lfloor \sqrt{a} \rfloor$ immediately to the left, or right, of k, using (9.2) with $L+2$. Finally, Markov's inequality applied to (9.1) puts every median within $C_b\sqrt{a}$ of a (enlarging C_b as needed).

9.4. **Endpoint hitting-time means.** Let T_j be the first hitting time of j and define the one-edge passage means

$$e_k = \mathbb{E}_k T_{k+1} = \frac{F_k}{p(k)\lambda_k}, \qquad f_k = \mathbb{E}_k T_{k-1} = \frac{G_k}{p(k)\mu_k}.$$

These standard identities also follow by summing the Poisson equation $Qh = -1$ against p up to the absorbing edge. The recurrences are

$$\lambda_k e_k - \mu_k e_{k-1} = 1, \qquad \mu_k f_k - \lambda_k f_{k+1} = 1,$$

with $e_1 = 1/\lambda_1$ and $f_n = 1/\mu_n$. Since λ_k decreases and μ_k increases, induction in the first recurrence shows that e_k increases with k; reverse induction shows that f_k decreases with k.

Fix a stationary median m, and choose a fixed L so large that $|m-a| \le (L-2)\sqrt{a}$. Put $u = \lfloor a - L\sqrt{a} \rfloor$ and $v = \lceil a + L\sqrt{a} \rceil$. The stationary estimates imply, uniformly on the central interval $[u, v]$,

$$e_k + f_k \le C_{b,L}\sqrt{a}$$

whenever the respective passage is defined: indeed $p(k) \ge c_{b,L}/\sqrt{a}$ and $\lambda_k \wedge \mu_k \ge c_b > 0$ there. Hence all passages between u, m, and v together contribute $O_{b,L}(a)$ to their mean times.

For $k \le u$ write $\delta_k = \lambda_k - \mu_k = -D_k > 0$ and $h_k = 1/\delta_k$. The recurrence gives $e_k \le h_k$ by induction, since h_k increases. If $\Delta_k = h_k - e_k$ and $q_k = \mu_k/\lambda_k$, then

$$\Delta_1 = 0, \qquad \Delta_k = q_k \Delta_{k-1} + q_k(h_k - h_{k-1}).$$

The sequence q_k increases, and

$$h_k - h_{k-1} = \frac{2k + b - 2}{n\delta_k\delta_{k-1}}$$

also increases. Iterating and bounding the resulting geometric series therefore gives

$$0 \le h_k - e_k \le \frac{q_k}{1-q_k}(h_k - h_{k-1}) \le \frac{\mu_k(\delta_{k-1} - \delta_k)}{\delta_k^3} \le \frac{C_b a^2}{(a-k)^3}.$$

The last inequality uses $n \asymp_b a^2$, $\mu_k \le C_b$, $\delta_{k-1} - \delta_k \le C_b/a$, and $\delta_k \ge c_b(a-k)/a$. Summing for $k \le u$ yields

$$\sum_{k=1}^{u} e_k = \sum_{k=1}^{u} \frac{1}{\delta_k} + O_{b,L}(a),$$

because $\sum_{d \ge L\sqrt{a}-1} d^{-3} = O_L(a^{-1})$.

For $k \geq v$, put $h_k = 1/D_k$. This time h_k decreases, and $h_k - h_{k+1}$ decreases as well: the function $x \mapsto n/[x^2 + (b-1)x - bn]$ is decreasing and convex for $x > a$ (and n large). The ratio $p_k^* = \lambda_k/\mu_k$ decreases. Reverse iteration of the recurrence, starting at $f_n = h_n$, gives

$$0 \leq h_k - f_k \leq \frac{p_k^*}{1-p_k^*}(h_k - h_{k+1}) \leq \frac{\lambda_k(D_{k+1} - D_k)}{D_k^3} \leq \frac{C_b a^2}{(k-a)^3}.$$

At $k = n$ the difference is zero. For the final estimate, use $\lambda_k \leq b$, $D_{k+1} - D_k = (2k+b)/n$, and $D_k \geq c_b(k-a)(k+a)/n$; hence the displayed ratio is at most $C_b n^2/[(k-a)^3(k+a)^2] \leq C_b a^2/(k-a)^3$. It follows that

$$\sum_{k=v}^{n} f_k = \sum_{k=v}^{n} \frac{1}{D_k} + O_{b,L}(a).$$

The reciprocal drift sums have elementary partial fractions:

$$\sum_{k=1}^{u} \frac{1}{\delta_k} = \frac{n}{2a+b-1} \sum_{k=1}^{u} \left(\frac{1}{a-k} + \frac{1}{a+k+b-1} \right),$$
$$\sum_{k=v}^{n} \frac{1}{D_k} = \frac{n}{2a+b-1} \sum_{k=v}^{n} \left(\frac{1}{k-a} - \frac{1}{k+a+b-1} \right).$$

Integral comparison for the decreasing functions $1/x$ shows that each bracketed sum is $\frac{1}{2}\log a + O_{b,L}(1)$: the first singular sum runs from a distance of order a down to $L\sqrt{a}$, and in the second sum the upper endpoint contributes only $\log[(n-a)/(n+a+b-1)] = O_b(a/n)$ after cancellation. Thus, by the strong Markov property and addition of passage means,

$$\boxed{\begin{aligned} \mathbb{E}_i T_m &= \frac{n}{4a}\log a + O_b(a) \\ &= \frac{\sqrt{n}\log n}{8\sqrt{b}} + O_b(\sqrt{n}), \qquad i \in \{1, n\}. \end{aligned}}$$

More explicitly, both endpoint means separately equal the last expression. The same proof works for any target j with $|j - a| \leq L_0\sqrt{a}$, with constants allowed to depend on the fixed L_0.

9.5. **Endpoint hitting-time variances.** Successive one-edge passage times are independent by the strong Markov property. We prove

$$\operatorname{Var}_1 T_m + \operatorname{Var}_n T_m = O_b(a^2) = O_b(n).$$

For an upward passage $T = T_{k+1}$ started at k, let $H(j) = \mathbb{E}_j T_{k+1} = \sum_{\ell=j}^{k} e_\ell$ for $j \leq k$. The equations $QH = -1$ and $Q\mathbb{E}_\cdot T_{k+1}^2 = -2H$ on $\{1, \ldots, k\}$, with zero boundary value at $k+1$, give upon summing against p

$$\begin{aligned} \mathbb{E}_k T_{k+1}^2 &= \frac{2}{p(k)\lambda_k} \sum_{j=1}^{k} p(j)H(j) \\ &= \frac{2}{p(k)\lambda_k} \sum_{\ell=1}^{k} e_\ell F_\ell \leq 2e_k^2 \sum_{\ell=1}^{k} \frac{F_\ell}{F_k}. \end{aligned}$$

The inequality uses monotonicity of e_ℓ and $e_k = F_k/[p(k)\lambda_k]$. In particular,

$$\operatorname{Var}_k T_{k+1} \leq 2e_k^2 \sum_{\ell=1}^{k} \frac{F_\ell}{F_k}. \tag{9.4}$$

The reflected argument, using monotonicity of f_k, gives

$$\operatorname{Var}_k T_{k-1} \leq 2f_k^2 \sum_{\ell=k}^{n} \frac{G_\ell}{G_k}. \tag{9.5}$$

For $k \le u = \lfloor a - L\sqrt{a} \rfloor$, set

$$r_k = \frac{p(k-1)}{p(k)} = \frac{\mu_k}{\lambda_{k-1}} \le \frac{\mu_k}{\lambda_k} < 1,$$

with $r_1 = 0$. These ratios increase with k. Shifting every index in the sum defining F_ℓ by $k - \ell$ gives $F_\ell \le r_k^{k-\ell} F_k$. Consequently,

$$\sum_{\ell=1}^{k} \frac{F_\ell}{F_k} \le \frac{1}{1 - r_k} \le \frac{\lambda_k}{\delta_k}.$$

Together with $e_k \le 1/\delta_k$, this implies

$$\operatorname{Var}_k T_{k+1} \le \frac{2\lambda_k}{\delta_k^3} \le \frac{C_b a^3}{(a-k)^3}.$$

Summing over $k \le u$ gives $O_b(a^2)$.

For $k \ge v = \lceil a + L\sqrt{a} \rceil$, use instead the decreasing ratios

$$s_k = \frac{p(k+1)}{p(k)} = \frac{\lambda_k}{\mu_{k+1}} \le \frac{\lambda_k}{\mu_k} < 1,$$

with $s_n = 0$. The same index-shifting argument gives

$$\sum_{\ell=k}^{n} \frac{G_\ell}{G_k} \le \frac{1}{1 - s_k} \le \frac{\mu_k}{D_k}.$$

Since $f_k \le 1/D_k$,

$$\operatorname{Var}_k T_{k-1} \le \frac{2\mu_k}{D_k^3}.$$

For $v \le k \le 2a$, this is at most $C_b a^3/(k-a)^3$, whose sum is $O_b(a^2)$. For $k > 2a$, use $D_k \ge c_b k^2/n$ and $\mu_k \le k^2/n$ to obtain the bound $C_b n^2/k^4$; its sum is $O_b(n^2/a^3) = O_b(a)$.

It remains to treat the $O_b(\sqrt{a})$ central passages. Uniformly for $k \in [u, v]$, the stationary estimates and the mean formulas give $e_k + f_k \le C_b\sqrt{a}$ and $F_k \wedge G_k \ge c_b > 0$. Furthermore,

$$\sum_{\ell=1}^{k} F_\ell = \mathbb{E}_p[(k - K + 1)_+] \le |k - a| + 1 + \mathbb{E}_p|K - a| \le C_b\sqrt{a},$$

$$\sum_{\ell=k}^{n} G_\ell = \mathbb{E}_p[(K - k + 1)_+] \le C_b\sqrt{a}.$$

Equations (9.4) and (9.5) therefore bound each central passage variance by $C_b a^{3/2}$. Their total is $O_b(a^2)$. Adding independent passage variances proves

$$\boxed{\operatorname{Var}_1 T_m \vee \operatorname{Var}_n T_m \le C_b a^2 \le C_b n.}$$

As with the means, this bound remains valid for every target within a fixed multiple of $\sqrt{a}$ of a, with the constant depending on that multiple.

9.6. **Application of the birth–death cutoff theorem.** We use Chen–Saloff-Coste [11, Theorem 1.4(1)]. In the continuous-time setting, if m_n has stationary mass bounded below on both sides, put

$$\theta_n = \max\{\mathbb{E}_1 T_{m_n}, \mathbb{E}_n T_{m_n}\}, \qquad \alpha_n^2 = \max\{\operatorname{Var}_1 T_{m_n}, \operatorname{Var}_n T_{m_n}\}.$$

Their theorem states that $\theta_n/\alpha_n \to \infty$ implies a worst-case total-variation (θ_n, α_n) cutoff. Its published normalization starts from a stochastic matrix B_n and uses the generator $B_n - I$. To apply it here, take

$$c_n \ge \max_k(\lambda_k + \mu_k), \qquad B_n = I + Q_n/c_n.$$

The continuous-time chain with generator $B_n - I$ is our chain slowed by c_n. All passage means and standard deviations are multiplied by c_n; both the criterion and its time-window conclusion therefore transfer exactly to Q_n.

Take m_n to be any stationary median. The previous two subsections give

$$\theta_n = t_n + O_b(a), \qquad \alpha_n = O_b(a), \qquad \frac{\theta_n}{\alpha_n} \longrightarrow \infty.$$

The theorem yields

$$t^{\mathrm{count}}_{\mathrm{mix},n}(\varepsilon) = t_n + O_{b,\varepsilon}(a) = t_n + O_{b,\varepsilon}(\sqrt{n}).$$

This proves the worst-case count assertion of Theorem 9.1.

To prove the assertion for each endpoint separately, it remains only to give endpoint lower bounds; their upper bounds follow from the worst-case bound. For a fixed $L > 0$, let $j_- = \lfloor a - L\sqrt{a} \rfloor$ and $j_+ = \lceil a + L\sqrt{a} \rceil$. The stationary moment estimate gives

$$p([1, j_- - 1]) \vee p([j_+ + 1, n]) \le C_b/L.$$

The preceding passage estimates, now with these central targets, give

$$|\mathbb{E}_1 T_{j_-} - t_n| + |\mathbb{E}_n T_{j_+} - t_n| \le C_{b,L} a, \qquad \mathrm{Var}_1 T_{j_-} \vee \mathrm{Var}_n T_{j_+} \le C_{b,L} a^2.$$

By nearest-neighbor paths, $\{T_{j_-} > t\} \subseteq \{K_t < j_-\}$ when $K_0 = 1$, and $\{T_{j_+} > t\} \subseteq \{K_t > j_+\}$ when $K_0 = n$. Chebyshev's inequality therefore gives, for $A > C_{b,L}$ and n large,

$$d_{n,1}(t_n - Aa) \wedge d_{n,n}(t_n - Aa) \ge 1 - \frac{C_b}{L} - \frac{C_{b,L}}{(A - C_{b,L})^2}.$$

First choose L and then A large. Together with the count upper bound, this proves both endpoint cutoffs with an $O_b(\sqrt{n})$ window. The exact endpoint identity in the first subsection completes the polymer endpoint claims.

9.7. **A further consequence: separation cutoff.** Define separation from an initial state x by

$$\mathrm{sep}_x(t) = 1 - \min_y \frac{P_t(x,y)}{\pi(y)}.$$

The same established passage estimates give

$$s_n := \mathbb{E}_1 T_m + \mathbb{E}_n T_m = \frac{\sqrt{n}\log n}{4\sqrt{b}} + O_b(\sqrt{n}), \qquad v_n := \sqrt{\mathrm{Var}_1 T_m + \mathrm{Var}_n T_m} = O_b(\sqrt{n}).$$

Chen–Saloff-Coste [11, Theorem 1.3(1)] implies endpoint separation cutoff at s_n with window v_n. In a continuous-time birth–death chain maximum separation is attained from either boundary state; see their Remark 1.2. Thus

$$t^{\mathrm{count}}_{\mathrm{sep},n}(\varepsilon) = \frac{\sqrt{n}\log n}{4\sqrt{b}} + O_{b,\varepsilon}(\sqrt{n}).$$

The singleton-fiber identity gives a density that is constant on every count fiber. Taking the minimum of that density, rather than summing its absolute difference from one, proves the exact endpoint separation identity for the polymer chain as well. Both polymer endpoint separation mixing times therefore have this same asymptotic formula. This again does not yield an arbitrary-start polymer upper bound.

10. Mixing from an arbitrary polymer configuration

Theorem 10.1. *For every fixed $b > 0$, the polymer chain with $\beta = b/n$ satisfies*

$$t_{\mathrm{mix},n}^{\mathrm{polymer}}(\varepsilon) \le C_b\sqrt{n}\left\{(\log n)^2 + \log(1/\varepsilon)\right\}, \qquad 0 < \varepsilon < 1,$$

for all sufficiently large n. Together with endpoint cutoff this gives, for each fixed $\varepsilon \in (0,1)$,

$$\frac{\sqrt{n}\log n}{8\sqrt{b}} - O_{b,\varepsilon}(\sqrt{n}) \le t_{\mathrm{mix},n}^{\mathrm{polymer}}(\varepsilon) \le O_b(\sqrt{n}(\log n)^2).$$

The extra logarithm in the upper bound is not removed by this argument; worst-case polymer cutoff is not asserted.

10.1. **A degree-dependent sector bound.** Let L_B be the labelled-cut comparison generator, and let $\mathcal{L}_\ell$ be its degree-ℓ radial generator:

$$(\mathcal{L}_\ell g)(k) = \beta(n-k-\ell)(g(k+1)-g(k)) + \frac{(k-1)(k-\ell)}{n}(g(k-1)-g(k)) - \left(\ell\beta + \frac{\ell(k-1)}{n}\right)g(k), \qquad \ell \le k \le n-\ell.$$

Assume $2 \le A := n\sqrt{\beta} \le n/4$. For $\ell \ge 1$ use

$$h_\ell(k) = \frac{1}{(k+A)_\ell}, \qquad (x)_\ell = x(x+1)\cdots(x+\ell-1).$$

Its neighboring ratios are

$$\frac{h_\ell(k+1)}{h_\ell(k)} = \frac{k+A}{k+A+\ell}, \qquad \frac{h_\ell(k-1)}{h_\ell(k)} = \frac{k+A+\ell-1}{k+A-1}.$$

Direct substitution gives the exact identity

$$\frac{-(\mathcal{L}_\ell h_\ell)(k)}{h_\ell(k)} = \ell\left[\frac{\beta(n-k-\ell)}{k+A+\ell} + \beta + \frac{(k-1)(A+\ell-1)}{n(k+A-1)}\right]. \tag{10.1}$$

All bracketed terms are nonnegative. If $k \le A$, then $\ell \le k$ gives $n-k-\ell \ge n/2$ and $k+A+\ell \le 3A$; the first term is at least $\sqrt{\beta}/6$. If $k \ge A$, then

$$\frac{k-1}{k+A-1} \ge \frac{1}{3}, \qquad A+\ell-1 \ge A,$$

so the last term is at least $\sqrt{\beta}/3$. The finite positive-vector comparison therefore proves that every eigenvalue in degree ℓ is at least

$$\lambda_\ell \ge \frac{\ell\sqrt{\beta}}{6}. \tag{10.2}$$

Independently, the killing term gives

$$\lambda_\ell \ge \ell\beta + \frac{\ell(\ell-1)}{n}.$$

In particular, at fixed $b > 0$, $\beta = b/n$, and $A = \sqrt{bn}$,

$$\lambda_\ell \ge \frac{b\ell}{6A} \qquad (1 \le \ell \le \lfloor n/2 \rfloor)$$

for every sufficiently large n.

10.2. **A polynomial bound on the count density.** Fix $b > 0$, put $A = \sqrt{bn}$, and let p be the stationary count law. Let r be a mode of p and define

$$V(k) = \left(\frac{p(r)}{p(k)}\right)^{3/4} \geq 1, \qquad W(k) = (\sqrt{\lambda_k} - \sqrt{\mu_k})^2.$$

We prove the drift inequality

$$\frac{(QV)(k)}{V(k)} \leq \frac{C_b - c_b \log V(k)}{A}, \qquad 1 \leq k \leq n, \tag{10.3}$$

for all sufficiently large n.

By detailed balance, the exact expression is

$$\frac{QV(k)}{V(k)} = \lambda_k^{1/4}\mu_{k+1}^{3/4} + \mu_k^{1/4}\lambda_{k-1}^{3/4} - \lambda_k - \mu_k.$$

At an endpoint the term with zero outgoing rate is zero; thus the displayed formula remains valid if one formally sets $\lambda_0 = b$ and $\mu_{n+1} = n + 1$. These are algebraic conventions only. Write $\lambda = \lambda_k$ and $\mu = \mu_k$. With $u = \lambda^{1/4}$ and $v = \mu^{1/4}$,

$$\begin{aligned}\lambda + \mu - \lambda^{1/4}\mu^{3/4} - \mu^{1/4}\lambda^{3/4} \\ = (u - v)^2(u^2 + uv + v^2) \geq \frac{3}{4}(u - v)^2(u + v)^2 = \frac{3}{4}W(k).\end{aligned}$$

The two coefficient-variation errors are

$$E_1 = \lambda_k^{1/4}(\mu_{k+1}^{3/4} - \mu_k^{3/4}), \qquad E_2 = \mu_k^{1/4}(\lambda_{k-1}^{3/4} - \lambda_k^{3/4}).$$

Since $\mu_{k+1} - \mu_k = 2k/n$, concavity gives, for $k \geq 2$,

$$E_1 \leq \frac{3}{4}\lambda_k^{1/4}\mu_k^{-1/4}\frac{2k}{n} \leq C_b n^{-3/4}\sqrt{k}.$$

The same bound holds at $k = 1$ by direct substitution, since $E_1 = \lambda_1^{1/4}(2/n)^{3/4}$. For $k \leq 2A$, it follows that $E_1 \leq C_b/A$. For $k > 2A$, one has $\mu_k \geq 3b \geq 3\lambda_k$ and hence $W(k) \geq c_b k^2/n$ (for large n). Thus

$$\frac{E_1}{W(k)} \leq C_b n^{1/4} k^{-3/2} \leq \frac{C_b}{A}.$$

For sufficiently large n, this is at most $1/4$. The inequality $(x + y)^{3/4} - x^{3/4} \leq y^{3/4}$ also gives, uniformly including $k = n$,

$$E_2 \leq \mu_k^{1/4}(b/n)^{3/4} \leq \frac{b^{3/4}\sqrt{k}}{n} \leq \frac{C_b}{A}.$$

Consequently

$$\frac{QV(k)}{V(k)} \leq -\frac{1}{2}W(k) + \frac{C_b}{A}. \tag{10.4}$$

It remains to check

$$\log V(k) \leq C_b\{1 + AW(k)\}. \tag{10.5}$$

On $A/2 \leq k \leq 2A$, the elementary stationary ratio estimates proved earlier give

$$\log \frac{p(r)}{p(k)} \leq C_b\left(1 + \frac{(k - A)^2}{A}\right).$$

The exact drift factorization around its zero $a = A + O_b(1)$ and $\lambda_k + \mu_k \leq C_b$ give $W(k) \geq c_b(k - a)^2/A^2$, proving (10.5) on this interval. If $k < A/2$, then $W(k) \geq c_b > 0$, while

$$\log \frac{p(r)}{p(k)} \leq \log \frac{p(r)}{p(1)} \leq \log \frac{A^{2(r-1)}}{(r - 1)!\, r!} \leq C_b A,$$

where $r = A + O_b(1)$ and the last bound follows from $j! \ge (j/e)^j$. If $k > 2A$, sum the consecutive ratios to obtain

$$\log \frac{p(r)}{p(k)} = \sum_{j=r}^{k-1} \left[\log \frac{j}{A} + \log \frac{j+1}{A} - \log(1 - j/n) \right] \le \frac{C_b k^2}{A}.$$

Here $\log x \le x$ controls the first two terms. For the last, $-\log(1 - j/n) \le 2j/n$ when $k \le n/2$; when $k > n/2$, its sum is at most $\sum_{j=1}^{n-1} -\log(1 - j/n) \le Cn \le Ck^2/n$. Since $W(k) \ge c_b k^2/n$ and $A^2 = bn$, this proves (10.5) in the remaining case. Combining it with (10.4) proves (10.3).

Now let $y(t) = \mathbb{E}_i V(K_t)$ for any initial count i. The finite-state evolution equation and Jensen's inequality for $x \log x$ imply

$$y'(t) \le \frac{C_b}{A} y(t) - \frac{c_b}{A} y(t) \log y(t).$$

Thus $z(t) = \log y(t)$ satisfies

$$z(t) \le \frac{C_b}{c_b} + e^{-c_b t/A} z(0).$$

Uniformly in i, (10.5) and $W(i) \le \lambda_i + \mu_i \le b + n$ give $z(0) \le C_b n^2$. Consequently there exists $C_b < \infty$ such that

$$\sup_i \sup_{t \ge t_0} \mathbb{E}_i V(K_t) \le C_b, \qquad t_0 = C_b A \log n. \tag{10.6}$$

Let q_t be any count distribution at time $t \ge t_0$ started from a point mass (or any mixture). Equation (10.6) gives $q_t(k) \le C_b / V(k)$. The stationary Gaussian and geometric-tail estimates give

$$p(r) \asymp_b A^{-1/2}, \qquad \sum_k \sqrt{p(k)} \le C_b A^{1/4}.$$

It follows that

$$M(t) := \sum_{k=1}^{n} \frac{q_t(k)^2}{p(k)} \le C_b p(r)^{-3/2} \sum_k \sqrt{p(k)} \le C_b A, \qquad t \ge t_0. \tag{10.7}$$

This is a polynomial bound, not a claim that count L^2 mixing is complete.

10.3. **The polymer mixing bound.** Let p_{cut} be the map from nonempty labelled cut sets to their unordered segment lengths, and write π_B, π_M, and p for the stationary cut, polymer, and count laws, respectively. Start the actual polymer chain from any state x. Let f_t be its density with respect to π_M, and lift it by

$$F_t(S) = f_t(p_{\mathrm{cut}}(S)).$$

The lift is nonnegative. Its average on the layer $|S| = k$ is

$$\overline{F}_t(k) = \frac{q_t(k)}{p(k)},$$

where q_t is the distribution of the actual polymer count. Strong lumpability of the count means that q_t evolves under Q from $K(x)$. The preceding Lyapunov calculation therefore gives, with $A = \sqrt{bn}$,

$$M(t) := \sum_k p(k) \overline{F}_t(k)^2 \le M_* := C_b A, \qquad t \ge t_0 := C_b A \log n. \tag{10.8}$$

Choose the constant so that $M_* \ge 1$.

Let P_ℓ denote the orthogonal projection onto the complete degree-ℓ sector of the labelled-cut comparison space, and set

$$d_\ell = \dim H_\ell = \binom{n}{\ell} - \binom{n}{\ell - 1}.$$

On a layer of size $N_k = \binom{n}{k}$, the projector onto degree ℓ has constant diagonal d_ℓ/N_k whenever that degree is present. Indeed, the projector commutes with every permutation of the labels, and its trace is d_ℓ. Positive semidefiniteness then gives

$$|P_\ell(S,T)| \le \sqrt{P_\ell(S,S)P_\ell(T,T)} = d_\ell/N_k.$$

For every nonnegative function F, it follows, using the ordinary matrix kernel of this layerwise projector, that

$$\begin{aligned}\|P_\ell F\|^2_{L^2(\pi_B)} &= \sum_k \frac{p(k)}{N_k} \sum_{|S|=|T|=k} F(S)P_\ell(S,T)F(T)\\ &\le d_\ell \sum_k p(k)\overline{F}(k)^2.\end{aligned}$$

Absent degrees contribute zero. Telescoping the dimensions gives

$$\sum_{\ell=0}^{d} \|P_\ell F_t\|_2^2 \le \binom{n}{d} M(t), \qquad 0 \le d \le \lfloor n/2 \rfloor. \tag{10.9}$$

Put $R(t) = \|F_t\|^2_{L^2(\pi_B)} = \|f_t\|^2_{L^2(\pi_M)}$ and $g = \sqrt{\beta}/6 = b/(6A)$. The stationary Dirichlet-form identity and the proved sector bound $\lambda_\ell \ge g\ell$ imply

$$\begin{aligned}-R'(t) &= 2\mathcal{E}_M(f_t,f_t) = 2\mathcal{E}_B(F_t,F_t)\\ &\ge 2g(d+1)\sum_{\ell>d}\|P_\ell F_t\|_2^2\\ &\ge 2g(d+1)\left\{R(t) - M_*\binom{n}{d}\right\}, \qquad t \ge t_0.\end{aligned}$$

This is a statement about the actual polymer evolution; no invariance of the polymer lift under the cut semigroup, or under individual sector projections, is assumed. The sector decomposition is used only to bound the instantaneous stationary energy.

Positivity also bounds the initial size of this differential inequality: on each layer, $\sum_S F_t(S)^2 \le (\sum_S F_t(S))^2$, so

$$R(t) \le \sum_k p(k)\binom{n}{k}\overline{F}_t(k)^2 \le 2^n M_*, \qquad t \ge t_0.$$

For n large, whenever $R(t) > 16M_*n^2$, put

$$s(t) = \log\frac{R(t)}{M_*}, \qquad d = \left\lfloor \frac{s(t)}{2\log n} \right\rfloor.$$

Then $1 \le d < \lfloor n/2 \rfloor$, because $s(t) \le n\log 2$. Moreover,

$$M_*\binom{n}{d} \le M_* n^d \le M_* e^{s(t)/2} \le R(t)/4.$$

The energy inequality consequently gives

$$s'(t) = \frac{R'(t)}{R(t)} \le -g(d+1) \le -\frac{g}{2\log n}s(t).$$

Since $s(t_0) \le n\log 2$, it takes at most $2g^{-1}(\log n)^2$ additional time to reach $R(t) \le 16M_*n^2$. Once this threshold is reached it remains satisfied, because $R(t)$ is nonincreasing.

Finally use the already established polymer gap $\gamma_M \ge c_b/A$. If t_1 is the preceding threshold time, then

$$R(t_1+u) - 1 \le e^{-2\gamma_M u}\{R(t_1)-1\}, \qquad \|P^M_{t_1+u}(x,\cdot) - \pi_M\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{R(t_1+u)-1}.$$

As $R(t_1) \le 16 C_b A n^2$, an additional $C_b A\{\log n + \log(1/\varepsilon)\}$ time suffices. This proves Theorem 10.1, uniformly over the initial polymer state x.

11. Cut reactions with graph diffusion

Let $G = (V, E, c)$ be a connected weighted graph on $n \ge 2$ vertices, with symmetric nonnegative edge weights c_{xy}. Its random-walk Laplacian is

$$(\Delta_G a)(x) = \sum_y c_{xy}[a(x) - a(y)],$$

and $\lambda_G > 0$ denotes its smallest eigenvalue on the mean-zero subspace. On the nonempty subsets of V, define

$$\begin{aligned}(\mathcal{L}_{\beta,D,G} f)(S) &= \beta \sum_{x \notin S} [f(S \cup \{x\}) - f(S)] \\ &\quad + \frac{|S| - 1}{n} \sum_{x \in S} [f(S \setminus \{x\}) - f(S)] \\ &\quad + D \sum_{\{x,y\} \in E} c_{xy}[f(S^{xy}) - f(S)], \qquad \beta > 0, \quad D \ge 0,\end{aligned} \tag{11.1}$$

where S^{xy} exchanges the occupancy of x and y. The deletion sum is zero at $|S| = 1$. The first two terms are the cut-reaction generator, and the last term is conservative exclusion along G. The reversible probability measure remains

$$\pi_\beta(S) = Z^{-1} \frac{(\beta n)^{|S|}}{(|S| - 1)!}. \tag{11.2}$$

Indeed, its adjacent-cardinality weights satisfy detailed balance for the reaction rates, and it is invariant under every coordinate permutation.

The cardinality is the birth–death chain with rates $\beta(n - k)$ and $k(k - 1)/n$; write $\gamma_Q(n, \beta)$ for its gap. The degree-one radial reaction operator acts on $k = 1, \dots, n - 1$ as

$$\begin{aligned}(A_1 h)(k) &= \beta(n - k - 1)[h(k + 1) - h(k)] \\ &\quad + \frac{(k - 1)^2}{n}[h(k - 1) - h(k)] - \left(\beta + \frac{k - 1}{n}\right) h(k).\end{aligned} \tag{11.3}$$

Let $\eta_1(n, \beta)$ be the smallest eigenvalue of $-A_1$.

Theorem 11.1 (Exact gap with graph diffusion)**.** *For every n, β, D, G as above, the spectral gap of (11.1) is*

$$\gamma_{\mathrm{sp}}(n, \beta, D, G) = \min\{\gamma_Q(n, \beta),\ \eta_1(n, \beta) + D\lambda_G\}. \tag{11.4}$$

Proof. Use the Boolean harmonic decomposition of the cut chain. In degree ℓ, functions are sums of products $h(|S|)H(S)$, where H belongs to the degree-ℓ harmonic space and h is supported on the allowable cardinalities. The reaction generator acts on this sector as $A_\ell \otimes I$, where $A_\ell = \mathcal{L}_\ell$. Every coordinate permutation preserves harmonic degree and commutes with the canonical lifts between slices. Hence the graph-swap generator acts as $I \otimes \mathcal{S}_{G,\ell}$, with the same harmonic action at every cardinality. The slice inner products on the harmonic factor differ only by a scalar, so these are tensor decompositions of the reversible Hilbert spaces as well. Consequently the generator on the degree-ℓ sector is

$$A_\ell \otimes I + I \otimes D\mathcal{S}_{G,\ell}. \tag{11.5}$$

The negative of the graph-swap operator is nonnegative, since it is the restriction of a reversible Markov generator with its sign changed.

In degree zero, swaps act trivially and the first positive eigenvalue is γ_Q. In every degree $\ell \geq 2$, Proposition 7.2 gives $\inf \mathrm{Spec}(-A_\ell) \geq \gamma_Q$; adding the graph form cannot lower this bound.

For degree one write

$$H_a(S) = \sum_{x \in S} a(x), \qquad \sum_{x \in V} a(x) = 0.$$

A direct calculation gives $\mathcal{S}_G H_a = -H_{\Delta_G a}$. Thus the degree-one eigenvalues are sums of eigenvalues of $-A_1$ and of $D\Delta_G$ on mean-zero vectors. Their smallest value is $\eta_1 + D\lambda_G$. Taking the minimum over all degrees proves (11.4). □

Only the one-particle graph action is required in this proof. Higher harmonic sectors already lie above the count gap before diffusion is added.

Proposition 11.2 (An additive-error formula for the degree-one gap)**.** *Set*

$$d = \sqrt{\beta^2 + 4\beta}, \qquad c_1 = \frac{\beta + d}{2}, \qquad a = nc_1.$$

Then

$$c_1 - \frac{2a}{n(a+2)} \leq \eta_1(n, \beta) \leq c_1 - \frac{2a^2}{n(n+a-2)(n+a)}. \tag{11.6}$$

In particular,

$$c_1 - \frac{2}{n} \leq \eta_1(n, \beta) \leq c_1. \tag{11.7}$$

Proof. The number a is positive and satisfies $a^2 = \beta n(n+a)$. For $h(k) = 1/(k+a)$ the quotient in (11.13) simplifies to

$$-\frac{A_1 h(k)}{h(k)} = \frac{a}{n} - \frac{2a^2}{n(k+a-1)(k+a+1)}. \tag{11.8}$$

Every denominator is positive, even when $a < 1$, since $k \geq 1$. The quotient increases with k. Its values at $k = 1$ and $k = n-1$ give (11.6) by positive-vector comparison. The simpler bracket follows from $a/(a+2) \leq 1$. □

Corollary 11.3 (Leading constants with graph diffusion)**.** *With d and c_1 as above, put*

$$m = \min\{d, c_1 + D\lambda_G\}.$$

The count-chain bracket $d - 5/n \leq \gamma_Q \leq d + 4/n$ from Theorem 7.4 and Theorem 11.1 imply

$$m - \frac{5}{n} \leq \gamma_{\mathrm{sp}}(n, \beta, D, G) \leq m + \frac{4}{n}. \tag{11.9}$$

In particular, if $n^2\beta \longrightarrow \infty$, the ratio of γ_{sp} to m tends to one, uniformly in the graph and in $D \geq 0$.

Proof. Combine Theorem 7.4 and the degree-one additive-error bracket inside the minimum in (11.4). For the final assertion, observe that $d \geq 2\sqrt{\beta}$ and $c_1 \geq \sqrt{\beta}$, so $nm \geq n\sqrt{\beta} \longrightarrow \infty$. □

Proposition 11.4 (The degree-one scale)**.** *For every $n \geq 2$ and $\beta > 0$, set*

$$u = \beta + \min\{n\beta, \sqrt{\beta}\}.$$

Then

$$\frac{u}{8} \leq \eta_1(n, \beta) \leq u. \tag{11.10}$$

More precisely, with $c_0 = 1 - 2^{-1/2}$,

$$\max\left\{\beta, \min\left(c_0 n\beta, \frac{1}{2n}\right)\right\} \leq \eta_1 \leq \min\left\{\frac{n\beta}{2}, \beta + \sqrt{\beta}\right\}. \tag{11.11}$$

Also,

$$\eta_1 \le \gamma_Q - \frac{2}{n}. \tag{11.12}$$

In particular, the gap of the cut-reaction chain without graph diffusion is exactly η_1.

Proof. The positive test function $h(k) = 1/k$ gives the exact quotient

$$-\frac{A_1 h(k)}{h(k)} = \frac{\beta n}{k+1} \le \frac{n\beta}{2},$$

which proves the first upper bound. For the second, put $j = k-1$ and consider the conservative part of A_1. Its birth rate is $\beta(n-2-j)$, its death rate is j^2/n, and its killing is $\beta + j/n$. If ν is the invariant law of this finite conservative chain, stationarity applied to j yields

$$\mathbb{E}_\nu j^2 = \beta n(n-2-\mathbb{E}_\nu j) \le \beta n^2.$$

The constant-function Rayleigh quotient, followed by Cauchy–Schwarz, therefore gives $\eta_1 \le \beta + \sqrt{\beta}$. When $n = 2$ the conservative chain has the single state $j = 0$, so the same argument applies.

The killing term gives $\eta_1 \ge \beta$. For another lower bound use $h(k) = k^{-1/2}$. At $k = 1$ its quotient is

$$\beta + \beta(n-2)(1-2^{-1/2}) \ge c_0 n\beta.$$

For $k \ge 2$, discard the nonnegative birth term and the term β to obtain

$$-\frac{A_1 h(k)}{h(k)} \ge \frac{k-1}{n}\left[1-(k-1)\left(\sqrt{\frac{k}{k-1}}-1\right)\right] \ge \frac{1}{2n}.$$

Here $(k-1)(\sqrt{k/(k-1)}-1) = 1/(\sqrt{k/(k-1)}+1) \le 1/2$. Positive-vector comparison proves (11.11).

It remains to obtain a square-root lower bound in the intermediate range. If $n \ge 8$ and $4/n^2 \le \beta \le 1/16$, put $a = n\sqrt{\beta}$, so that $2 \le a \le n/4$, and use $h(k) = 1/(k+a)$. Direct cancellation gives

$$-\frac{A_1 h(k)}{h(k)} = \beta + \frac{\beta(n-k-1)}{k+a+1} + \frac{a(k-1)}{n(k+a-1)}. \tag{11.13}$$

If $k \le a$, then $n-k-1 \ge n/2$ and $k+a+1 \le 5a/2$, so the second term is at least $\sqrt{\beta}/5$. If $k \ge a$, then $k-1 \ge k/2$ and $k+a-1 \le 2k$, so the final term is at least $\sqrt{\beta}/4$. Thus $\eta_1 \ge \sqrt{\beta}/5$ in this range.

The estimates patch with a universal constant. For $\beta \le n^{-2}$, the lower bound $c_0 n\beta$ is at least $(n+1)\beta/8 = u/8$. For $n \le 7$ and $\beta \ge n^{-2}$, $u \le 8\beta$. If $n \ge 8$ and $n^{-2} \le \beta \le 4n^{-2}$, the auxiliary bound is at least $c_0/n \ge \sqrt{\beta}/7$; combine this with $\eta_1 \ge \beta$. In the intermediate range just treated the two lower bounds give $u \le 6\eta_1$. Finally, for $\beta \ge 1/16$, $u \le 5\beta$. These cases prove the lower bound in (11.10). The two upper bounds in (11.11) immediately give $\eta_1 \le u$.

Finally, let h be the positive ground state of the count-derivative operator $A_{n,\beta}$. Its forward ratios are at least $1/2$ by Lemma 8.2 (or by the weaker ratio bound used in Proposition 7.2). Direct subtraction gives

$$\frac{(A_{n,\beta}-A_1)h(k)}{h(k)} = \frac{(k-1)h(k-1)/h(k) - 2k}{n} \le -\frac{2}{n}.$$

At $k = 1$ the backward term is zero. Testing $-A_1$ with h proves (11.12), and Theorem 11.1 gives the last assertion. □

Corollary 11.5 (A uniform spatial phase diagram)**.** *Put* $s = \beta + \sqrt{\beta} + n^{-1}$ *and* $u = \beta + \min\{n\beta, \sqrt{\beta}\}$. *Then*

$$\frac{1}{8}\min\{s, u + D\lambda_G\} \le \gamma_{\mathrm{sp}}(n,\beta,D,G) \le 4\min\{s, u + D\lambda_G\}. \tag{11.14}$$

In particular, for $0 < \beta \le n^{-2}$,

$$\gamma_{\mathrm{sp}}(n, \beta, D, G) \asymp \min\{n^{-1}, n\beta + D\lambda_G\}, \tag{11.15}$$

with absolute comparison constants.

Proof. Combine Theorem 11.1 and Proposition 11.4 with the count-chain bounds $s/6 \le \gamma_Q \le 4s$. In the stated rare-fragmentation range, $s \asymp n^{-1}$ and $u \asymp n\beta$. □

For a unit-rate cycle on $n \ge 3$ vertices, $\lambda_G = 4\sin^2(\pi/n)$. For the discrete torus of side $L \ge 3$ in dimension d, with unit rates along its edges, $n = L^d$ and $\lambda_G = 4\sin^2(\pi/L)$. Thus (11.15) becomes

$$\gamma_{\mathrm{sp}} \asymp \min\{n^{-1}, n\beta + D/L^2\}$$

on a torus. This identifies the competition between count relaxation, reactions that renew a dilute cut configuration, and graph diffusion.

On a cycle, adding or deleting a cut splits or merges neighboring intervals, and exchanging adjacent occupied and unoccupied vertices moves an interval boundary. On a general graph, (11.1) is an exclusion process with cardinality-dependent creation and deletion. Its reaction rates retain the global count dependence and its equilibrium law remains exchangeable. These features permit the tensor decomposition and specify the scope of the spatial result.

11.1. **Inhomogeneous reversible reactions.** The exact formula also controls reaction rates that depend on the positions and surrounding configuration, through a comparison of reversible conductances. For each undirected reaction edge $e = \{S, S \cup \{x\}\}$ with $S \ne \varnothing$ and $x \notin S$, choose a factor a_e satisfying

$$0 < a_- \le a_e \le a_+ < \infty.$$

Multiply both the addition rate from S to $S \cup \{x\}$ and the reverse deletion rate by this same factor. Keep the graph-swap generator and its diffusion coefficient D unchanged. Write γ_{inhom} for the resulting gap.

Corollary 11.6 (Bounded inhomogeneous reaction rates)**.** *The modified chain is reversible for* (11.2) *and satisfies*

$$\begin{aligned} \min\{a_-\gamma_Q,\ a_-\eta_1 + D\lambda_G\} &\le \gamma_{\mathrm{inhom}} \\ &\le \min\{a_+\gamma_Q,\ a_+\eta_1 + D\lambda_G\}. \end{aligned} \tag{11.16}$$

The factors may depend arbitrarily on the edge and its configuration; the modified reaction generator need not commute with permutations.

Proof. Multiplication by the same factor in both directions preserves detailed balance at each reaction edge. Let $\mathcal{E}_B$ be the reaction form and $\mathcal{E}_G$ the unit-diffusion graph-swap form, both with respect to π_β. The modified form satisfies

$$a_-\mathcal{E}_B(f,f) + D\mathcal{E}_G(f,f) \le \mathcal{E}_{\mathrm{inhom}}(f,f) \le a_+\mathcal{E}_B(f,f) + D\mathcal{E}_G(f,f).$$

For any constant $a > 0$, Theorem 11.1, with diffusion coefficient D/a and then a change of time by a, gives the gap of $aB + D\mathcal{S}_G$ as $\min\{a\gamma_Q, a\eta_1 + D\lambda_G\}$. All three forms have the same variance denominator. Taking their infima proves (11.16). □

For example, place the vertices in a cyclic order and let adding a cut at x divide an interval into two positive lengths i and j. For $|\varepsilon| < 1/2$, set

$$a_{\{S,S\cup\{x\}\}} = 1 + \varepsilon\cos\left(\frac{2\pi x}{n}\right) + \varepsilon\frac{(i-j)^2}{(i+j)^2}. \tag{11.17}$$

This combines a spatially varying environment with a dependence on the neighboring interval lengths. The factor is unchanged when i and j are interchanged, and the same split determines it for the reverse merger. It satisfies $1 - 2|\varepsilon| \le a_e \le 1 + 2|\varepsilon|$, so Corollary 11.6 applies with these

constants. The corollary covers bounded reversible modifications of the stated reaction edges; its proof does not extend the exact formula to arbitrary coagulation or fragmentation kernels.

12. Nonreversible conservative motion of cuts

The reduction in Theorem 11.1 extends to conservative jumps that permute the sites, without requiring inverse jumps to have the same rate. The state in this section is the full nonempty cut set. As in the spatial model, unordered interval lengths need not form a Markov process.

For $\sigma \in S_n$ put $(T_\sigma F)(S) = F(\sigma S)$, and let

$$H = \sum_{\sigma \in S_n} r_\sigma (T_\sigma - I), \qquad r_\sigma \ge 0, \qquad L = L_B + DH, \quad D \ge 0. \tag{12.1}$$

Terms with $r_\sigma = 0$ may be omitted. No relation between r_σ and $r_{\sigma^{-1}}$ is imposed. The induced one-coordinate generator is

$$(H_1 a)(x) = \sum_\sigma r_\sigma (a(\sigma x) - a(x)). \tag{12.2}$$

It preserves the subspace $\mathbb{C}_0^n = \{a : \sum_x a(x) = 0\}$. Adjoints below are with respect to the uniform measure on coordinates or π_B on cut sets, as appropriate. Define

$$\kappa(H_1) = \min\{\operatorname{Re} z : z \in \operatorname{Spec}(-H_1|_{\mathbb{C}_0^n})\}, \tag{12.3}$$

$$\kappa_{\mathrm{s}}(H_1) = \lambda_{\min}\Big(-\tfrac{1}{2}(H_1 + H_1^*)|_{\mathbb{C}_0^n}\Big). \tag{12.4}$$

Zeros are included in these minima. Thus the definitions also apply when the coordinate motion alone has several communicating classes.

Theorem 12.1. *For every $n \ge 2$, $\beta > 0$, $D \ge 0$, and choice of the rates in (12.1), the law π_B in (2.2) is invariant for L. On the mean-zero subspace, the asymptotic spectral rate is exactly*

$$g_{\mathrm{as}} := \min\{\operatorname{Re} z : z \in \operatorname{Spec}(-L),\ z \ne 0\} = \min\{\gamma_Q,\ \eta_1 + D\kappa(H_1)\}. \tag{12.5}$$

The largest constant g_{s} for which

$$\|e^{tL}F\|_{L^2(\pi_B)} \le e^{-g_{\mathrm{s}} t}\|F\|_{L^2(\pi_B)} \qquad (t \ge 0,\ \mathbb{E}_{\pi_B} F = 0) \tag{12.6}$$

is

$$g_{\mathrm{s}} = \min\{\gamma_Q,\ \eta_1 + D\kappa_{\mathrm{s}}(H_1)\}. \tag{12.7}$$

Here η_1 is the smallest eigenvalue of $-\mathcal{L}_1$, as in Theorem 11.1.

Proof. Every coordinate permutation preserves π_B, which is constant on each cardinality layer. Hence H preserves π_B, and $T_\sigma^* = T_{\sigma^{-1}}$. The reaction generator is irreducible, so adding the nonnegative jump rates DH preserves irreducibility. In particular, zero is the simple stationary eigenvalue of L.

The Boolean harmonic decomposition is equivariant under every coordinate permutation. On degree ℓ, the reaction generator acts on the radial factor, and H acts on the harmonic factor. Thus, with H_ℓ denoting the latter action,

$$L|_\ell = \mathcal{L}_\ell \otimes I + I \otimes DH_\ell. \tag{12.8}$$

The inner product factors into the radial reversible inner product and the harmonic inner product. Each permutation is unitary on the harmonic factor, so for every complex vector v in that factor,

$$\operatorname{Re}\langle v, -H_\ell v\rangle = \frac{1}{2}\sum_\sigma r_\sigma \|T_\sigma v - v\|^2 \ge 0.$$

Consequently every eigenvalue of $-H_\ell$ has nonnegative real part.

Diagonalizing the self-adjoint radial factor in (12.8) expresses its spectrum as the sums of radial eigenvalues and D times eigenvalues of $-H_\ell$. This statement does not require diagonalizability of

H_ℓ. In degree zero, $H_0 = 0$, and the first positive eigenvalue is γ_Q. By Proposition 7.2, all reaction eigenvalues in degrees $\ell \geq 2$ are at least γ_Q. These sectors therefore cannot lower the minimum real part.

In degree one, H_ℓ is precisely the coordinate operator (12.2) on $\mathbb{C}_0^n$: if $F_a(S) = \sum_{x\in S} a(x)$, then $HF_a = F_{H_1 a}$. The smallest radial eigenvalue is η_1. Both terms on the right of (12.5) are attained by eigenvalues, proving that formula.

For the contraction statement, the symmetric part is $L_{\mathrm{s}} = L_B + D(H + H^*)/2$. It is a reversible permutation-jump generator plus L_B. The same sector argument gives its spectral gap as (12.7). For $F_t = e^{tL}F$,

$$\frac{d}{dt}\|F_t\|_2^2 = 2\,\mathrm{Re}\langle F_t, LF_t\rangle = 2\langle F_t, L_{\mathrm{s}}F_t\rangle \leq -2g_{\mathrm{s}}\|F_t\|_2^2.$$

Integration proves (12.6). Conversely, differentiating any proposed prefactor-one bound at $t = 0$ gives the Poincaré inequality for $-L_{\mathrm{s}}$. Its gap is therefore the largest possible such constant. □

Remark 12.2 (Asymptotic rate and finite-time norm bounds). In finite dimension,

$$\lim_{t\to\infty} -\frac{1}{t}\log\|e^{tL}|_{L_0^2(\pi_B)}\| = g_{\mathrm{as}}.$$

For every $\varepsilon > 0$ there is a finite C_ε such that the norm is at most $C_\varepsilon e^{-(g_{\mathrm{as}}-\varepsilon)t}$. If a defective eigenvalue lies on the line with real part $-g_{\mathrm{as}}$, a polynomial factor in t can occur. Thus (12.5) alone does not assert a prefactor-one estimate at that exponent. Formula (12.7) identifies the exact exponent for that stronger type of estimate.

12.1. **An explicit strict increase of the asymptotic rate.** On three coordinates, take

$$H = \tfrac{1}{2}(T_{(12)} - I) + \tfrac{1}{2}(T_{(23)} - I) + (T_{(123)} - I).$$

The convention $(123) : 1 \mapsto 2 \mapsto 3 \mapsto 1$ gives

$$H_1 = \begin{pmatrix} -3/2 & 3/2 & 0 \\ 1/2 & -2 & 3/2 \\ 1 & 1/2 & -3/2 \end{pmatrix}. \tag{12.9}$$

Its rows and columns sum to zero, but it is not symmetric. The characteristic polynomials of $-H_1$ and its symmetric part are

$$z(z^2 - 5z + 27/4), \qquad z(z-2)(z-3),$$

respectively. Hence

$$\kappa(H_1) = \frac{5}{2}, \qquad \kappa_{\mathrm{s}}(H_1) = 2, \qquad \mathrm{Spec}(-H_1) = \{0,\, 5/2 + i/\sqrt{2},\, 5/2 - i/\sqrt{2}\}.$$

The two rates in Theorem 12.1 become

$$g_{\mathrm{as}} = \min\{\gamma_Q, \eta_1 + 5D/2\}, \qquad g_{\mathrm{s}} = \min\{\gamma_Q, \eta_1 + 2D\}.$$

For $0 < D < 2(\gamma_Q - \eta_1)/5$, the asymptotic rate is strictly larger than the reversible symmetric-part rate.

The same example extends to arbitrarily many sites. Let $V = \{1,2,3\} \times W$, where W carries a connected symmetric random-walk generator G with gap at least $5/2$. At each $w \in W$, apply the three permutations above inside $\{1,2,3\} \times \{w\}$ with the displayed rates. For every graph edge wz and every $i \in \{1,2,3\}$, also transpose (i,w) and (i,z) at that edge's rate. The one-coordinate generator is then

$$H_1^V = H_1 \otimes I + I \otimes G.$$

Its eigenvalues add, so its two relevant rates remain $5/2$ and 2. The preceding exact formulas therefore hold for every $n = 3|W|$, with γ_Q and η_1 evaluated at that n. All conservative moves in this construction affect two or three sites.

Remark 12.3 (Scope of the conservative extension). The rates in (12.1) specify unconditional permutation jumps; an unchanged occupancy configuration simply contributes zero. Occupancy-dependent directed exclusion rates are not covered by this argument. In that model, the action on a linear coordinate observable generally contains quadratic occupancy terms, and the degree-one tensor reduction used here need not hold.

12.2. **A stationary circulation in the polymer reaction graph.** There is also a nonreversible extension on the original unlabelled polymer state space. It uses only the existing merge and split edges. The modification is state dependent; it is not a kernel depending only on the sizes of the reacting clusters. Antisymmetric perturbations preserving equilibrium are an established mechanism for accelerating the asymptotic convergence rate; see, for example, Hwang, Hwang-Ma and Sheu [19] for the diffusion setting. The construction below verifies that mechanism on an explicit cycle of the present polymer reaction graph.

Lemma 12.4. *For every $n \geq 2$ and $\beta > 0$, the eigenvalue $\gamma_M = \gamma_Q$ of $-L_M$ is simple. Its real eigenfunction can be written $g(K(m))$, with $g(k+1) - g(k) > 0$ for $1 \leq k < n$ after a choice of sign.*

Proof. The positive ground state h of the count-derivative operator is the discrete derivative of a first nonconstant count eigenfunction, so the latter is strictly monotone. That count eigenvalue is simple. We check that no higher sector can tie it.

The minimum-ratio argument in Lemma 7.1 gives the strict inequality $h(i+1)/h(i) > 1/2$. Indeed, if a minimum ratio $r \leq 1/2$ occurred at i, the two adjacent eigenvalue equations would give

$$0 \geq \beta(1-r) + \frac{2i}{n}(r^{-1} - 1) - \frac{2}{n} > 0.$$

At $r = 1/2$ the last expression is $\beta/2 + 2(i-1)/n > 0$; smaller r only increases it. The endpoint conventions are the same as in that proposition. In its higher-sector comparison the extra term $(\ell - 1)\beta[2h(r+1)/h(r) - 1]$ is consequently strictly positive for every retained degree $\ell \geq 2$. Thus all these sectors have bottom eigenvalue strictly above γ_Q. A polymer observable has no degree-one component. Equality in its Poincaré inequality therefore forces its lift to lie in the one-dimensional first nonconstant count eigenspace. For $n = 2, 3$ there are no retained higher sectors to check. □

For $n \geq 4$, consider the four distinct polymer configurations

$$x = (2, 1^{n-2}), \qquad u = (2, 2, 1^{n-4}), \qquad v = (4, 1^{n-4}), \qquad w = (3, 1^{n-3}),$$

written as multisets of cluster sizes. Every edge of the directed cycle $x \to u \to v \to w \to x$ is an allowed reaction. Let $J(a, b)$ be 1 on its forward edges, -1 on its reverse edges, and 0 elsewhere. For an oriented edge a, b, write $C(a, b) = \pi_M(a)q(a, b) = \pi_M(b)q(b, a)$. Choose

$$0 < |\varepsilon| < \min\{C(x, u), C(u, v), C(v, w), C(w, x)\} \tag{12.10}$$

and define

$$q_\varepsilon(a, b) = q(a, b) + \varepsilon \frac{J(a, b)}{\pi_M(a)} \quad (a \neq b). \tag{12.11}$$

Since $\sum_b J(a, b) = 0$, the diagonal generator entries are unchanged.

Proposition 12.5. *The rates (12.11) define an irreducible nonreversible polymer chain with invariant law π_M. Its symmetric part is exactly L_M. Its asymptotic spectral rate is strictly greater than γ_M, whereas its best prefactor-one $L^2(\pi_M)$ contraction exponent is exactly γ_M.*

Proof. Condition (12.10) keeps every original reaction rate positive. The antisymmetry and zero divergence of J show that π_M is stationary. The stationary forward and backward fluxes on each cycle edge differ by 2ε, so detailed balance fails. Write $L_\varepsilon = L_M + \varepsilon A$, where $A(a, b) = J(a, b)/\pi_M(a)$ for $a \neq b$ and $A(a, a) = 0$. Then $A1 = 0$ and $A^* = -A$, proving the symmetric-part claim.

Let $g(K)$ be the eigenfunction in Lemma 12.4, normalized in $L^2(\pi_M)$. The two states u, w have cardinality $n-2$, while x, v have cardinalities $n-1, n-3$. Consequently

$$(Ag)(u) = \frac{g(n-3) - g(n-1)}{\pi_M(u)} \neq 0.$$

Also $\langle g, Ag\rangle_{\pi_M} = 0$ by skew-adjointness.

For a nonzero eigenvalue λ of $-L_\varepsilon$ with complex eigenvector F, stationarity implies $\mathbb{E}_{\pi_M} F = 0$. Therefore

$$\operatorname{Re}\lambda = \frac{\langle F, -L_M F\rangle_{\pi_M}}{\|F\|_2^2} \geq \gamma_M.$$

Equality would put F in the complex span of g, by simplicity of the bottom eigenspace. But $-L_\varepsilon g = \gamma_M g - \varepsilon Ag$ is not in that span. Equality is impossible. Since the spectrum is finite, its least nonzero real part is strictly larger than γ_M.

The symmetric part remains L_M, so differentiation of the squared L^2 norm gives contraction at exponent γ_M. Differentiating at $t = 0$ with $F = g$ rules out any larger exponent with prefactor one. □

The small-circulation improvement has a positive explicit second-order coefficient. Put $B = -L_M$, let g have norm one, and let R be the inverse of $B - \gamma_M I$ on $\{1, g\}^\perp$. The lowest nonzero eigenvalue branch satisfies

$$\lambda(\varepsilon) = \gamma_M + \varepsilon^2 \langle Ag, R\,Ag\rangle_{\pi_M} + O(\varepsilon^4), \qquad \langle Ag, R\,Ag\rangle_{\pi_M} > 0. \tag{12.12}$$

Indeed, simple-eigenvalue perturbation gives first correction $R\,Ag$ to the eigenvector and the displayed quadratic coefficient. The branch is real near zero. Since $L_{-\varepsilon} = L_\varepsilon^*$, that real branch is even in ε, giving the stated remainder. The coefficient is positive because $Ag \neq 0$ and R is positive definite on $\{1, g\}^\perp$. The separated spectrum at $\varepsilon = 0$ ensures that this branch is the asymptotic rate for sufficiently small ε.

This construction proves strict acceleration for each fixed n, β. It does not provide an improvement bounded away from zero uniformly in n: the admissible stationary current in (12.10) and the coefficient in (12.12) depend on the chosen cycle.

13. Reversible changes of the coagulation kernel

The stationary law can be preserved while allowing the reaction rates to depend on cluster sizes. Fix positive symmetric multipliers $c_n(i,j) = c_n(j,i)$ for $i + j \leq n$. An unordered pair of cluster instances of sizes i, j merges at rate $2c_n(i,j)/n$. Each internal link whose removal produces lengths i, j fragments at rate $\beta c_n(i,j)$. Thus, for $i < j$, the coarse forward and reverse rates are

$$\frac{2c_n(i,j)}{n} m_i m_j, \qquad 2\beta c_n(i,j) m_{i+j},$$

whereas for $i = j$ they are

$$\frac{c_n(i,i)}{n} m_i(m_i - 1), \qquad \beta c_n(i,i) m_{2i}.$$

Theorem 13.1. *The modified chain is reversible for the original law π_M in (2.3). If*

$$0 < c_- \leq c_n(i,j) \leq c_+ < \infty \qquad (i + j \leq n),$$

then its gap γ_c satisfies

$$c_- \gamma_M(n,\beta) \leq \gamma_c(n,\beta) \leq c_+ \gamma_M(n,\beta). \tag{13.1}$$

In particular, with $s = \beta + \sqrt{\beta} + n^{-1}$,

$$\frac{c_-}{6} s \leq \gamma_c \leq 4c_+ s.$$

The comparison constants are uniform in n, β whenever c_-, c_+ are.

Proof. Each reaction and its reverse have been multiplied by the same positive number. Every original detailed-balance identity therefore remains valid. Writing the Dirichlet form as a sum over undirected reaction edges gives

$$c_-\mathcal{E}_M(f,f) \le \mathcal{E}_c(f,f) \le c_+\mathcal{E}_M(f,f).$$

The variance uses the same stationary law for all three forms. Taking infima of the Rayleigh quotients proves (13.1). □

This is a robustness consequence of Dirichlet-form comparison. Its proof also applies to symmetric multipliers attached to full undirected state-space edges. It does not establish the sharp scale of an unbounded kernel from its typical value under equilibrium.

The stationary cut representation survives without a boundedness assumption on c_n. At a boundary between segments of lengths i, j, replace the deletion rate by $(k-1)c_n(i,j)/n$. At an internal link producing i, j, use the birth rate $\beta c_n(i,j)$. The law π_B is unchanged, and averaging over the uniform cyclic order conditional on the segment multiset gives the modified polymer generator. Consequently the exact variance and Dirichlet identities in (2.4) continue to hold.

The full harmonic reduction generally does not survive. For example, take $n = 4$ and $c(i,j) = i+j$. Cut configurations with segment lengths $(1,3)$ and $(2,2)$ both have $k = 2$, but their generator applied to k is, respectively, $6\beta - 2$ and $4\beta - 2$. Even the subspace of functions of cardinality is therefore not invariant for this modified cut generator.

13.1. **A classical multiplicative-kernel comparison class.** A different Gibbs equilibrium gives a useful growing-kernel baseline. Consider pair-merger rates $2ij/n$ and per-link fragmentation rate $\beta(i+j)$ when the two resulting lengths are i, j. Writing $\theta = \beta n$, the reversible law is

$$\pi_{\mathrm{E}}(m) = Z_{\mathrm{E}}^{-1}\frac{\theta^{K(m)}}{\prod_{j=1}^{n} j^{m_j} m_j!}. \tag{13.2}$$

This is the Ewens cycle-count law. The chain is the classical Metropolis transposition process on cycle types, with a deterministic change of time.

Proposition 13.2 (Classical spectral formula in the present units)**.** *For this multiplicative-kernel chain,*

$$\gamma_{\mathrm{E}}(n,\beta) = 2\beta + \frac{2(n-1)}{n}. \tag{13.3}$$

Multiplying both directions of each size-pair reaction by $c_n(i,j) \in [c_-, c_+]$ preserves (13.2) and gives

$$c_-\left(2\beta + \frac{2(n-1)}{n}\right) \le \gamma_{\mathrm{E},c} \le c_+\left(2\beta + \frac{2(n-1)}{n}\right).$$

Proof. Detailed balance in (13.2) follows directly from the weight ratio $w_i w_j / w_{i+j} = (i+j)/(ij)$ for $w_j = 1/j$. For unequal fragments the coarse split rate is $2\beta(i+j)m_{i+j}$, and for equal fragments it is $\beta(2i)m_{2i}$.

We record the normalization of the known spectrum explicitly. In Jiang [21, Eq. (7) and Theorem 4.2], the parameter of the Metropolis transition matrix is $t = \theta^{-1}$. Our continuous-time generator is $(n-1)\max\{1,\theta\}(P_t - I)$. Indeed, multiplying the uniform transposition proposal rate $\binom{n}{2}^{-1}$ by this factor gives merger rate $2/n$ per pair of labels in distinct cycles and split rate $2\theta/n$ per pair of labels in one cycle. Jiang's spectral formula consequently gives, for each partition $\lambda \vdash n$,

$$\Lambda_\lambda = \frac{2}{n}\left[\binom{n}{2} - \sum_i \binom{\lambda_i}{2} + \theta\sum_i (i-1)\lambda_i\right].$$

The zero eigenvalue corresponds to $\lambda = (n)$. For every other partition, $\binom{n}{2} - \sum_i \binom{\lambda_i}{2} \ge n-1$ and $\sum_i (i-1)\lambda_i \ge 1$. Both are equalities for $\lambda = (n-1,1)$. This proves (13.3); the comparison statement follows by the preceding Dirichlet-form argument. □

The spectral formula in Proposition 13.2 is an application of an established diagonalization, not a new solution of the multiplicative-kernel model. Its perturbation class illustrates that rates growing like ij can have a uniform gap of order $1+\beta$ once the fragmentation rule and equilibrium law are specified together.

13.2. **A uniform upper bound for two growing kernels.** In this subsection the equilibrium is the original law π_M. Let γ_{prod} and γ_{add} denote the gaps obtained by using the shared multipliers $c(i,j)=ij$ and $c(i,j)=i+j$, respectively. Thus the product model has pair-merger rate $2ij/n$ and per-link fragmentation rate βij; its fragmentation rule and equilibrium differ from those of the classical Ewens model above.

Proposition 13.3. *For every $n\geq 2$ and $\beta>0$,*

$$\gamma_M \leq \gamma_{\mathrm{prod}} \leq 2\beta + \frac{2(n-1)}{n}, \tag{13.4}$$

$$2\gamma_M \leq \gamma_{\mathrm{add}} \leq 4\beta + \frac{4(n-1)}{n}. \tag{13.5}$$

In particular the growing kernels have gap bounded above by a constant times $1+\beta$, uniformly in n.

Proof. The lower bounds follow from $ij\geq 1$ and $i+j\geq 2$ by Dirichlet-form comparison. To prove the upper bounds, write $X=m_1$ for the number of monomers, $K=\sum_j m_j$, and $\theta=\beta n$. For the shared-product generator L_{prod}, direct counting gives

$$L_{\mathrm{prod}}X = 2\beta(n-K) - \frac{2(n-1)}{n}X. \tag{13.6}$$

Indeed a j-cluster with $j\geq 3$ creates monomers at total rate $2\beta(j-1)$. A dimer splits at rate β and creates two monomers, giving the same contribution $2\beta(j-1)$. Merging a monomer with every possible cluster, including another monomer with its double loss, contributes $-2(n-1)X/n$.

We claim

$$\mathrm{Cov}_{\pi_M}(X,K) \leq \mathrm{Var}_{\pi_M}(X). \tag{13.7}$$

Condition on $X=s$. The remaining mass is $N=n-s$ and has no parts of size one. If $N=0$ its cluster count is zero. The value $N=1$ is impossible under this conditioning. For $N\geq 2$, the unnormalized probabilities of its cluster count $J_N=j$ are

$$a_N(j) = \frac{\theta^j}{j!}\binom{N-j-1}{j-1}, \qquad 1\leq j\leq \lfloor N/2\rfloor. \tag{13.8}$$

To obtain this formula, sum $1/\prod_{r\geq 2} m_r!$ over nonmonomer partitions with j parts. Multiplying by $j!$ counts ordered compositions of N into j parts each at least two, of which there are $\binom{N-j-1}{j-1}$.

On the common support for $N, N+1$,

$$\frac{a_{N+1}(j)}{a_N(j)} = \frac{N-j}{N-2j+1},$$

which is increasing in j: its derivative, viewed as a real function, is $(N-1)/(N-2j+1)^2>0$. Any newly available support point for $N+1$ is at the upper end. Thus J_{N+1} stochastically dominates J_N. For completeness, an increasing likelihood ratio gives this dominance by the nonnegative covariance of any increasing test function with the likelihood ratio under the first distribution; mass at a newly available largest point only increases such expectations. Also $J_0=0$ is stochastically below every J_N with $N\geq 2$. It follows that the function

$$s \longmapsto \mathbb{E}[K-X \mid X=s] = \mathbb{E}J_{n-s}$$

is decreasing on the feasible conditioning values $s \in \{0, \dots, n-2, n\}$. The missing value $n-1$ causes no difficulty. For a real random variable and a decreasing function of that variable, covariance is nonpositive, as follows by taking an independent copy and expanding the product of the two differences. Consequently

$$\operatorname{Cov}(X, K) = \operatorname{Var}(X) + \operatorname{Cov}(X, \mathbb{E}[K - X \mid X]) \le \operatorname{Var}(X),$$

proving (13.7).

The monomer count is nonconstant under π_M: both the all-monomer and single-cluster states have positive mass. Using it in the Rayleigh quotient and applying (13.6) gives

$$\gamma_{\mathrm{prod}} \le \frac{-\operatorname{Cov}(X, L_{\mathrm{prod}} X)}{\operatorname{Var}(X)} = \frac{2(n-1)}{n} + 2\beta \frac{\operatorname{Cov}(X, K)}{\operatorname{Var}(X)} \le \frac{2(n-1)}{n} + 2\beta.$$

Finally $i + j \le 2ij$ for positive integers i, j. The two chains have the same stationary law, so $\mathcal{E}_{\mathrm{add}} \le 2\mathcal{E}_{\mathrm{prod}}$ and $\gamma_{\mathrm{add}} \le 2\gamma_{\mathrm{prod}}$. This proves the remaining upper bound. □

These estimates constrain the relaxation scale without reducing the growing-kernel dynamics to a count chain. At $\beta = b/n$ with fixed $b > 0$, the comparison lower bounds have order $n^{-1/2}$ while the upper bounds have constant order. A lower bound of constant order in that regime is not established by Proposition 13.3.

14. Consequences and remaining questions

14.1. **Relaxation, mixing, and the visible modes.** For the polymer semigroup $P_t = \exp(tL_M)$, the exact reduction gives

$$\|P_t f\|_{L^2(\pi_M)} \le e^{-\gamma_Q t} \|f\|_{L^2(\pi_M)}, \qquad \pi_M f = 0. \tag{14.1}$$

The exponent is optimal. A strictly monotone first eigenfunction of the count chain attains it. At stationarity,

$$|\operatorname{Cov}_{\pi_M}(f(M_0), f(M_t))| \le e^{-\gamma_Q t} \operatorname{Var}_{\pi_M}(f). \tag{14.2}$$

For $\beta = b/n$, relaxation therefore occurs on the scale $\sqrt{n}/(2\sqrt{b})$, while total-variation mixing from either extreme polymer configuration has the larger scale $\sqrt{n}\log n/(8\sqrt{b})$. The logarithm measures the time needed for the count to enter its equilibrium fluctuation range. Its equilibrium location is of order $\sqrt{n}$, its standard deviation is of order $n^{1/4}$, and its local restoring rate is asymptotic to $2\sqrt{b}/\sqrt{n}$.

The distinction between labelled and unlabelled states is necessary when fragmentation is rare. At $n = 2$, the polymer chain has only one dimer and two monomers, with gap $\beta + 1$. The labelled cut chain has an additional eigenfunction given by the difference of the two singleton indicators, with eigenvalue $-\beta$. Both singleton cuts represent the same dimer, so this function cannot be a polymer observable. In general, rotation invariance removes degree one, and the sector comparison proves that no remaining polymer mode is slower than the count mode.

With spatial diffusion the labels are part of the state. The degree-one mode is then visible, and the exact minimum $\min\{\gamma_Q, \eta_1 + D\lambda_G\}$ determines which mode controls relaxation. At $\beta = b/n$ its two reaction rates are asymptotic to $2\sqrt{b}/\sqrt{n}$ and $\sqrt{b}/\sqrt{n}$. At $\beta = b/n^2$ the corresponding rates are $\rho(b)/n$ and $\sigma(b)/n$. These formulas give both the crossover and the diffusion strength needed for count relaxation to become the limiting mechanism.

14.2. **The remaining mixing question.** The count gap and the polymer gap agree, but a spectral-gap identity does not by itself identify the worst-case mixing time; see [24]. From a specified state m, the usual estimate is

$$\|P_t(m, \cdot) - \pi_M\|_{\mathrm{TV}} \le \tfrac{1}{2}\sqrt{\pi_M(m)^{-1} - 1}\, e^{-\gamma_Q t}.$$

Its prefactor can be large. The singleton-fiber argument avoids that prefactor for the two extreme states, because their evolving density is exactly constant on count fibers. An arbitrary initial partition has additional within-fiber information. Theorem 10.1 controls that information uniformly and yields, for each fixed $\varepsilon \in (0,1)$,

$$\frac{\sqrt{n}\log n}{8\sqrt{b}} - O_{b,\varepsilon}(\sqrt{n}) \le t_{\mathrm{mix},n}^{\mathrm{polymer}}(\varepsilon) \le C_b\sqrt{n}(\log n)^2.$$

The main open mixing problem is to determine whether this additional information disappears by $\sqrt{n}\log n/(8\sqrt{b}) + O_b(\sqrt{n})$ uniformly over initial partitions, or whether some configurations mix later. The endpoint cutoff theorem supplies the lower bound with the exact constant. It does not identify the worst initial polymer state, prove a worst-case polymer cutoff, or determine a limiting cutoff profile. An optimal window lower bound is also a separate question.

14.3. **Beyond the solvable reaction structure.** The bounded-conductance comparisons allow spatial and size dependence while retaining a specified equilibrium law. Growing kernels require further estimates because a uniform comparison with the constant kernel loses powers of n. For the shared multipliers ij and $i+j$, the monomer test function gives upper bounds of order $1+\beta$ (Proposition 13.3). At $\beta = b/n$, the proved lower bounds still have order $n^{-1/2}$, leaving a gap between the two estimates. The classical Ewens model shows that the fragmentation rule matters as much as the merger kernel: its equilibrium weights and its spectral scale are different from those of the constant-kernel polymer chain.

The nonreversible results distinguish two convergence quantities. A directed current can improve the least nonzero real part of the spectrum while leaving the symmetric part, and hence the best prefactor-one L^2 exponent, unchanged. The four-state polymer construction proves strict acceleration for each finite system. It leaves open a choice of currents with a gain bounded away from zero as n grows, as well as a quantitative total-variation comparison at the cutoff scale.

Finally, the spatial model studied here has local conservative motion but globally count-dependent reactions. A general spatial coagulation–fragmentation system may have spatially varying concentrations, cluster diffusion depending on size, collision kernels depending on distance, and a nonexchangeable equilibrium. Those changes remove the tensor decomposition used in the graph formula. Establishing comparable estimates for such systems would require new control of their spatial and cluster-size fluctuations.

Appendix A. An independent coupling proof of the square-root bound

This appendix proves Proposition 5.1 for the cut chain without the harmonic decomposition. It provides a probabilistic verification of the square-root estimate. The constants in the resulting total-variation bounds may depend on n and β; only their exponential rate is used.

Assume $n \ge 8$, $4/n^2 \le \beta \le 1/16$, and put $a = n\sqrt{\beta} \in [2, n/4]$. Start two copies of the cut process from adjacent nonempty sets. Until they meet, the coupling has two stages. After meeting, run the copies identically.

Stage I: one set contains the other. Suppose $X \subset Y$, $Y \setminus X$ consists of one bond, and $|X| = k$. Add every common vacant bond to both sets at rate β. For each common occupied bond, remove it from both sets at rate $(k-1)/n$, and remove it from Y alone at rate $1/n$. At the unique bond in $Y \setminus X$, add it to X at rate β and remove it from Y at rate k/n.

The first marginal has birth rate β per vacant bond and death rate $(k-1)/n$ per occupied bond. The second has birth rate β per vacant bond and death rate k/n per occupied bond. Hence both marginals agree with (2.1). The last two events couple the copies; a removal from Y alone at a common bond starts Stage II. Until one of these exits, k increases at rate $\beta(n-k-1)$ and decreases at rate $k(k-1)/n$. The total exit rate is $\beta + 2k/n$. Thus the killed internal generator is exactly $\mathcal{A}$ in (3.5).

Stage II: equal sizes and one exclusive bond each. Write $|X| = |Y| = r+1$, so the intersection contains r bonds. Add each common vacant bond to both sets at rate β, and remove each common occupied bond from both at rate r/n. At rate β, simultaneously add to each set the bond exclusive to the other set, so both become their union. At rate r/n, simultaneously remove each set's exclusive bond, so both become their intersection. When $r = 0$, the latter event has rate zero and does not create the empty set.

Each simultaneous absorption event produces one jump in each marginal. Accordingly the total absorption rate is $\beta + r/n$. The internal coordinate $r \in \{0, \dots, n-2\}$ has killed generator

$$(\mathcal{B}h)(r) = \beta(n-r-2)(h(r+1) - h(r)) + \frac{r^2}{n}(h(r-1) - h(r)) - \left(\beta + \frac{r}{n}\right) h(r). \tag{A.1}$$

The marginal birth and death rates are again those of (2.1).

Positive certificates and coupling time. For both stages use $h(z) = 1/(z+a)$. For Stage I, (5.2) gives $-(\mathcal{A}h)(k)/h(k) \geq \sqrt{\beta}/5$. For Stage II, exact cancellation yields

$$\frac{-(\mathcal{B}h)(r)}{h(r)} = \frac{\beta(n-r-2)}{r+a+1} + \beta + \frac{r(a-1)}{n(r+a-1)}. \tag{A.2}$$

If $r \leq a$, then $n - r - 2 \geq n/2$ and $r + a + 1 \leq 5a/2$, so the first term is at least $\sqrt{\beta}/5$. If $r \geq a$, then $r/(r+a-1) \geq 1/2$ and $a - 1 \geq a/2$, so the last term is at least $a/(4n) = \sqrt{\beta}/4$. Consequently, with $c = \sqrt{\beta}/5$, both killed generators satisfy $Ah \leq -ch$.

For completeness, any finite matrix A with nonnegative off-diagonal entries generates a positivity-preserving semigroup: add a sufficiently large scalar multiple of the identity to make the whole matrix nonnegative and use the exponential series. If $h > 0$ and $Ah \leq -ch$, differentiation gives

$$\frac{d}{dt}(e^{t(A+cI)}h) = e^{t(A+cI)}(A+cI)h \leq 0,$$

and hence $e^{tA}h \leq e^{-ct}h$. The survival time T_i of each stage therefore satisfies, uniformly over its starting states,

$$\mathbb{P}(T_i > t) \leq C_i e^{-ct}, \qquad C_i = \frac{\max h}{\min h} < \infty.$$

For every $0 < \theta < c$, integration of this tail bound shows that

$$\mathbb{E}e^{\theta T_i} \leq 1 + \frac{\theta C_i}{c - \theta}.$$

The strong Markov property at entry to Stage II, using its uniform bound over initial states, gives a finite exponential moment at rate θ for the total coupling time. Stage II has duration zero if Stage I couples directly. Thus the total coupling time has tail $O(e^{-\theta t})$.

The coupling inequality gives this rate for total variation between chains started at any adjacent nonempty sets. Any two nonempty sets are connected by a finite path through their union, using single-bond additions and deletions and never visiting the empty set. The triangle inequality gives the same rate for any two initial states.

Finally, let f be a nonconstant eigenfunction of L_B with eigenvalue $-\lambda$, and choose x, y with $f(x) \neq f(y)$. Then

$$e^{-\lambda t}|f(x) - f(y)| = |\mathbb{E}_x f(B_t) - \mathbb{E}_y f(B_t)| \leq 2\|f\|_\infty \, \|P_t(x,\cdot) - P_t(y,\cdot)\|_{\mathrm{TV}}.$$

The established exponential bound forces $\lambda \geq \theta$. Since this holds for every $\theta < c$, it follows that $\gamma_B \geq \sqrt{\beta}/5$. Proposition 2.1 then gives the polymer bound.

Appendix B. Reproducible finite computations

The accompanying program `code/check_certificates.py` checks the finite constructions appearing in the proof. Its exact part uses rational arithmetic from the Python standard library. A separate numerical part, using NumPy, constructs the full cut-set and polymer generators directly and compares them with the independently constructed radial operators. The analytical arguments in the preceding sections establish the theorem for all parameters; the computations check formulas, boundary conventions, and concrete instances.

The program first verifies five polynomial cancellation identities, including the reciprocal certificates and the coefficient in the up–down tower induction. It then enumerates cut sets and integer partitions, calculates averaged transition rates on every partition fiber, checks detailed balance, and verifies the count derivative identity as a matrix equation. The two coupling stages are checked by comparing every constructed marginal rate with the defining cut-set generator. Table 1 records the coverage of the computation supplied with this paper.

Table 1. Finite verification coverage. A parameter case fixes both n and β; the exact checks also enumerate the relevant states or matrix entries.

Construction	Coverage
Polynomial cancellations	Five exact identities
Stationary projection	$2 \le n \le 8$, 21 parameter cases; 654 averaged-rate and 654 detailed-balance identities
Count derivative	$2 \le n \le 12$; 1,716 exact matrix-entry identities
Reciprocal inequalities	770 upper, 210 count-lower, 1,152 sector-lower, and 212 coupling-lower parameter points
Final parameter cases	93 combinations, including the regime boundaries
Coupling marginals	$2 \le n \le 7$; 1,482 Stage-I and 2,046 Stage-II state/parameter cases
Full cut-set spectrum	$2 \le n \le 8$, five fragmentation parameters for each n

The numerical parameters are $\beta \in \{10^{-8}, n^{-2}, n^{-1}, 1, 100\}$. The largest full cut-set generator has dimension $2^8 - 1 = 255$. Across the 35 spectral comparisons, the maximum absolute discrepancy between the direct cut-set spectrum and the combined sector spectra was 1.48×10^{-12}. After division by the larger of one and the spectral radius, the maximum discrepancy was 2.11×10^{-15}. The directly enumerated polymer gaps also satisfied the theorem's inequalities throughout this grid.

The supplied result file was produced with Python 3.12.14 and NumPy 2.3.5. Python 3.10 or later suffices for the exact checks. From the package root, run

```
python code/check_certificates.py
```

to write `results/verification_results.json`. The option `-skip-numerical` runs only the exact part. Floating-point results may differ in their last digits across numerical libraries.

B.1. **Checks for the extensions.** The program `code/check_extensions.py` checks the additional spectral formulas against generators constructed directly from the reaction rules. Its output is `results/extension_checks.json`. Seven polynomial identities are checked exactly, including the sharp count and degree-one reciprocal formulas and the two critical finite-size corrections. Table 2 gives the additional coverage.

The maximum scaled discrepancy was 5.38×10^{-15} for the polymer/count equality, 7.47×10^{-15} for the reversible spatial formula, and 5.06×10^{-15} for the nonreversible spectral and symmetric-rate formulas. Every tested sharp bound and growing-kernel bound was satisfied. Every tested nonzero circulation gave a strict asymptotic spectral improvement. These finite checks do not establish a uniform lower bound on the size of that improvement.

TABLE 2. Finite checks of the extended results. The matrix comparisons use floating-point eigenvalues; the algebraic, flux, and drift identities specified as exact use rational arithmetic.

Construction	Coverage
Polymer/count gap equality	45 parameter cases, $2 \le n \le 10$
Sharp count and degree-one bounds	46 distinct cases through $n = 1000$; 8,544 degree-one quotient evaluations
Critical scaling functions	50 finite-error checks at 25 parameter pairs; 30 truncation consistency checks
Reversible graph diffusion	68 direct matrices on weighted paths, cycles, and complete graphs, $2 \le n \le 7$
Nonreversible permutation motion	24 direct matrices for $n = 3, 6$, checking real-part and symmetric gaps
Polymer circulation	20 cases for $4 \le n \le 8$; 240 stationarity and 848 symmetric-flux identities checked exactly
Growing kernels	48 generators for $2 \le n \le 9$; 570 monomer-drift identities, 2,184 detailed-balance identities, and 24 covariance inequalities checked exactly

The critical comparisons retain the analytic errors $2b/(R+3)$ and $b/(R+2)$ from the truncations of ρ and σ. The finite eigenvalues themselves are ordinary floating-point computations, not certified interval eigenvalues. The largest Jacobi matrix in the consistency checks has dimension 513.

The plotting program `code/plot_scaling.py` produces Figure 1. Its data are written to `results/scaling_data.json`. It computes 37 finite count gaps for $8 \le n \le 32768$ and 121 critical-parameter values for $10^{-3} \le b \le 10^4$. The truncation choice in the caption makes the analytic tail widths at most 0.166% and 0.323% of the computed finite eigenvalues for ρ and σ, respectively. All three programs can be run from the repository root:

```
python code/check_certificates.py
python code/check_extensions.py
python code/plot_scaling.py
```

The source, PDF, programs, and data are available at https://github.com/yspennstate/linear-polymer-relaxation.

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The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. An earlier proof was developed with OpenAI Codex agents, checked by separate agents, and reviewed with Claude (Anthropic). In October 2026, OpenAI's GPT-6 Astra Pro, working in ChatGPT with collaborating agents, was used to extend the results, reconstruct and check the proofs, examine the literature, carry out finite computations, and prepare the manuscript. Separate agents examined the stationary comparison, the birth–death estimates, the harmonic decomposition, the coupling argument, and the relation to earlier work. Claude Opus 5.5 (Anthropic) audited the final manuscript, including the proofs, the citations, and the finite computations. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

https://github.com/yspennstate/linear-polymer-relaxation

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com